From Novel View Synthesis to Generative World Modeling: A Comprehensive Survey

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Abstract

Novel View Synthesis (NVS)—rendering a scene from arbitrary unseen viewpoints given reference observations—has undergone a paradigm shift from classical geometry-based reconstruction toward learning-driven generative approaches. This survey examines NVS not merely as a rendering technique but as **a key spatial capability, particularly in the visual-generative branch** of Generative World Models: systems that predict visual scene states under arbitrary conditions including viewpoint changes, temporal dynamics, and agent interactions.

We present a unified conceptual framework organising existing methods along a *conceptual map*—from static multi-view reconstruction to dynamic, interactive world generation—across three axes: (i) scene representation (implicit, explicit, hybrid), (ii) generative paradigm (per-scene optimisation, feed-forward inference, diffusion-based synthesis), and (iii) world-modelling capability (geometry, dynamics, controllability). We survey more than **180+ methods in detail, drawing on a broader corpus of 400+ works** spanning NeRF, 3D Gaussian Splatting, generalizable reconstruction, diffusion-based NVS, and 4-D world modeling, and propose a roadmap toward physically grounded, generalisable world models. The scope of this survey is the *visual-generative* branch of world modeling; it does not aim to comprehensively cover latent dynamics models, model-based reinforcement learning, or planning-centric world model architectures.

1 Introduction

1.1 The Fundamental Problem

Intelligence requires reasoning about the world beyond what is immediately visible. A self-driving vehicle must anticipate the scene around a blind corner; a robotic arm must predict how a grasped object will appear from the wrist camera after a planned motion; a surgeon must mentally reconstruct anatomy from a 2-D scan. In each case, the core operation is identical: **synthesising a plausible visual state at an unobserved vantage point**.

This is the problem of **Novel View Synthesis (NVS)**. Formally, given a set of reference observations $\{\mathbf{I}_i, \boldsymbol{\pi}_i\}_{i=1}^n$ with known camera poses $\boldsymbol{\pi}_i$, the goal is to render a photo-realistic image $\hat{\mathbf{I}}$ at an arbitrary target pose $\boldsymbol{\pi}^*$, faithfully preserving 3-D geometry, surface appearance, and lighting. The problem is inherently ill-posed when n is small or when the target pose is far from any reference: an infinite family of 3-D scenes is consistent with any finite set of 2-D projections. Resolving this ambiguity requires either dense capture (classical IBR) or a learned prior over the distribution of plausible scenes (modern neural and generative approaches).

While NVS was studied in the image-based rendering era [29, 94, 119, 121], the past six years have witnessed an extraordinary paradigm shift. NeRF [11] demonstrated that a shallow MLP can represent complex scenes with view-dependent appearance by differentiably volume-rendering a continuous radiance field, establishing that neural representations could faithfully encode scene geometry and appearance from posed photographs alone. 3DGS [10] subsequently showed that an explicit set of anisotropic Gaussians can render at real-time frame rates, resolving the speed bottleneck that had limited NeRF's practical applicability. Most recently, diffusion-based methods [13, 130, 215] have moved the frontier toward *generalisable, generative* synthesis that requires as few as a single image—or even a text description—as input, fundamentally decoupling scene understanding from dense observation.

1.2 From NVS to World Modeling

We argue that this progression reflects a deeper conceptual shift: **NVS can be viewed as a key spatial inference capability underlying Generative World Modeling**.

Scope. This survey focuses on the *visual-generative* branch of world modeling, where geometry, viewpoint consistency, and scene simulation are central. It does not aim to comprehensively cover latent dynamics models, model-based reinforcement learning, or planning-centric world model architectures. Within this scope, NVS captures geometric state prediction under observer motion, but does not by itself solve object-centric dynamics, action consequences, or physical causality—capabilities addressed as the frontier in Sections 8 and 11.

> **Working Definition (World Model).** A *Generative World Model* is a learned system that can predict the visual (and optionally physical) state of a scene under arbitrary conditions—including viewpoint changes, temporal evolution, and agent interactions—in a manner that is *geometrically consistent*, *physically plausible*, and *generalisable* to scenes and conditions outside the training distribution.

Under this definition, NVS is the special case in which the only varying condition is camera pose. A full world model must additionally handle temporal dynamics (D3), action conditioning (D4), and physical plausibility (D5). Crucially, *spatial consistency enforced by NVS is a necessary prerequisite*: a world model without geometric grounding is merely a video predictor—perceptually plausible in distribution but unreliable for tasks requiring precise spatial reasoning such as robot manipulation or navigation. This observation motivates the organizing perspective of this survey: the capabilities developed by the NVS community provide essential geometric building blocks for visual-generative world modeling, even as full world modeling additionally requires physical causality, object-centric dynamics, and action-conditioned planning that NVS alone does not address.

Recent systems are actively closing this gap. Genie 2 [123] generates interactive 3-D environments from a single image; Cosmos [9] trains world foundation models at internet scale; and ViewCrafter [215] with StreetCrafter [207] demonstrate that camera-controlled video generation is practical at high resolution. These developments suggest an emerging trend toward a more unified model that subsumes NVS, video generation, and world simulation into a single architecture trained at scale.

1.3 Relationship to Prior Surveys

Table 1 compares this work with the most closely related prior surveys. Tewari *et al.* [162] and Gao *et al.* [46] focus on NeRF representations without addressing world modelling. Wu *et al.* [185] and Fei *et al.* [43] cover 3DGS but not generative methods or 4-D. Chen *et al.* [26] cover NVS for autonomous driving but lack a world-modeling lens. To our knowledge, **no existing survey jointly covers all three paradigms (NeRF, 3DGS, generative) through the unifying lens of Generative World Modeling**, nor proposes a continuous-spectrum taxonomy from static reconstruction to 4-D interactive generation. Our work fills this gap by treating the field as a single coherent research trajectory rather than a collection of independently evolving sub-communities.

1.4 The Unified Spectrum

We organise all methods along a *conceptual map* defined by three axes (Figure 1):

Table 1: Comparison with related surveys. ✓ covered; ∼ partially; – not covered.

Survey	Year	NeRF	3DGS	Generative	Dynamic	Generalise	4D/WM	Unified
Tewari *et al.* [162]	2022	✓	–	∼	∼	∼	–	–
Gao *et al.* [46]	2022	✓	–	–	∼	–	–	–
Zhu *et al.* [247]	2023	∼	–	–	–	–	–	–
Wu *et al.* [185]	2024	–	✓	–	∼	–	–	–
Fei *et al.* [43]	2024	–	✓	∼	∼	–	–	–
Chen *et al.* [26]	2024	∼	∼	–	✓	–	∼	–
Zhang *et al.* [237]	2025	∼	∼	–	–	✓	–	–
He *et al.* [57]	2025	–	✓	∼	∼	–	–	–
Ours	2026	✓	✓	✓	✓	✓	✓	✓

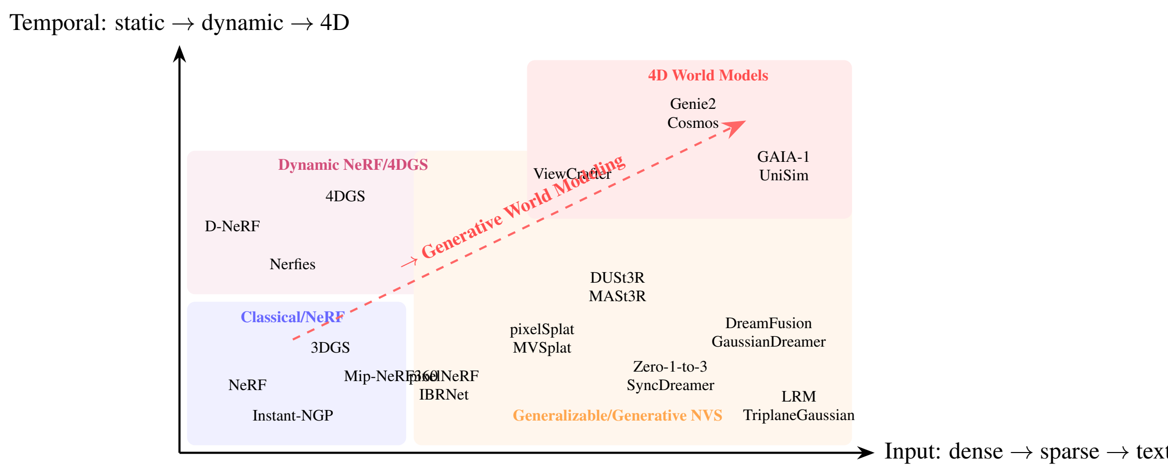

Figure 1: A conceptual map of the NVS landscape. The diagonal arrow marks the evolutionary trajectory toward 4-D Generative World Modeling.

Axis 1. Input regime: dense multi-view → sparse-view → single image → text/unconditional.

Axis 2. Temporal scope: static scene → deformable/dynamic → 4-D world generation.

Axis 3. Controllability: viewpoint only → viewpoint + time → viewpoint + time + action.

This conceptual map is an organizing lens rather than a claim about a single unified research trajectory: methods nearby in the figure need not be optimized for the same task, trained on the same data distribution, or evaluated under comparable protocols. Nevertheless, the map makes explicit which world-modeling capabilities each method addresses and which it leaves open. Methods near the origin of the spectrum—dense multi-view static reconstruction—are comparatively mature under controlled benchmarks, with current systems approaching photorealism within minutes of training. The open problems cluster at the far end: sparse-input, temporally extended, action-conditioned generation that generalises to arbitrary scenes. By mapping every method we survey to a position on this spectrum, we make explicit which aspects of the world-modeling desiderata each addresses and which it leaves open.

1.5 Contributions and Paper Structure

Contributions: (1) A unified continuous-spectrum taxonomy across three axes (input regime, temporal scope, controllability); (2) Cross-paradigm coverage of >180 methods with explicit World Modeling implications in each chapter; (3) A related-survey gap analysis (Table 1) substantiating the novelty of the World Modeling lens; (4) A five-direction open-problems roadmap.

The remainder is organised as follows. §2 establishes background and a unified formulation. §4–6 survey the three representation paradigms (NeRF, 3DGS, feed-forward reconstruction). §7 covers generative NVS with particular depth on camera-controlled diffusion. §8 examines 4-D generation, neural driving simulators, and world foundation models. §9–10 review datasets and evaluation protocols. §11 proposes a research roadmap.

2 Background and Unified Formulation

To appreciate the trajectory from classical image-based rendering to generative world modeling, one must first understand the foundational representations and rendering equations that all subsequent work builds upon. This section establishes the mathematical vocabulary and the key historical milestones that shaped the field. We begin with classical image-based rendering, which revealed the fundamental connection between captured images and 3-D light transport; we then introduce the two dominant neural representations, NeRF and 3DGS; and we conclude with a unified formulation that encompasses both NVS and world modeling as instances of a single conditional generation problem.

2.1 Classical Image-Based Rendering

Before the neural era, NVS was approached through *image-based rendering* (IBR) [29, 94, 119]. The foundational insight is that images directly encode light transport: sufficiently dense sampling of the plenoptic function [8] permits view interpolation without reasoning about explicit 3-D geometry. While this observation has proved remarkably durable—modern diffusion-based methods are, in a sense, learned plenoptic function approximators—the classical IBR literature also revealed the core bottleneck that motivated neural approaches: quality degrades sharply when the target viewpoint lies far outside the convex hull of captured cameras.

Light Field Rendering [94] and the Lumigraph [140] parameterise the light field on a two-plane slab, enabling photo-realistic view interpolation within the captured light-field volume. View interpolation [29] warps and blends reference images using depth estimates, anticipating modern depth-based view synthesis. Unstructured Lumigraphs and view-dependent texture mapping [121] relaxed capture requirements to unstructured camera sets, foreshadowing the transition toward casual capture settings. The geometry-estimation pipeline of Structure from Motion (SfM) [145] and Multi-View Stereo (MVS) [69] remains, to this day, the standard preprocessing step for all pose-conditioned NVS methods that require calibrated cameras.

The fundamental limitation of classical IBR is the requirement for *dense* capture: interpolation quality degrades rapidly as target viewpoints move beyond the convex hull of the observed camera set. Furthermore, these methods produce no compact scene model—each novel view requires querying the full set of reference images at inference time, making them ill-suited for real-time or storage-constrained applications. The desire for compact, generalisable scene representations drove the development of neural radiance fields, which we introduce next.

2.2 Volume Rendering and NeRF

NeRF [11] represents a scene as a continuous function $F_\theta : (\mathbf{x}, \mathbf{d}) \mapsto (\mathbf{c}, \sigma)$, where $\mathbf{x} \in \mathbb{R}^3$ is a 3-D location and $\mathbf{d} \in \mathbb{S}^2$ a viewing direction. The expected colour of a ray $\mathbf{r}(s) = \mathbf{o} + s\mathbf{d}$ is computed via classical volume rendering [81]:

$$\hat{C}(\mathbf{r}) = \int_{s_n}^{s_f} T(s)\,\sigma(\mathbf{r}(s))\,\mathbf{c}(\mathbf{r}(s), \mathbf{d})\,\mathrm{d}s, \quad T(s) = \exp\left(-\int_{s_n}^{s} \sigma(\mathbf{r}(u))\,\mathrm{d}u\right), \tag{1}$$

where $T(s)$ is the accumulated transmittance. F_θ is a position-encoded MLP [159] trained by minimising $\|\hat{C}(\mathbf{r}) - C(\mathbf{r})\|_2^2$ over all observed rays. Hierarchical stratified sampling decomposes the continuous integral into coarse and fine networks, concentrating samples near the visible surface.

NeRF's key properties—implicit representation, continuous differentiability, and view-dependent appearance via directional encoding—made it transformative. For the first time, a compact neural parameterisation could faithfully reproduce complex real scenes with fine specular highlights, transparency, and thin structures from a few dozen posed photographs, at a quality that previous methods required orders-of-magnitude more captures to achieve. Its central limitations are *slow per-scene optimisation* (hours) and *per-scene specificity* (no generalisation to unseen scenes), both of which the subsequent literature directly addressed through architectural redesign, explicit-hybrid representations, and large-scale pretraining.

2.3 Gaussian Splatting

3DGS [10] represents a scene as N anisotropic 3-D Gaussians $\{(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i, \alpha_i, \mathbf{c}_i)\}_{i=1}^N$, where $\boldsymbol{\mu}_i$ is the centre, $\boldsymbol{\Sigma}_i$ the full covariance, α_i the opacity, and $\mathbf{c}_i$ a set of spherical-harmonic coefficients encoding view-dependent colour. Novel views are rendered by projecting each Gaussian to a 2-D splat and compositing front-to-back:

$$\hat{C} = \sum_{i \in \mathcal{N}} \mathbf{c}_i \, \alpha_i \, \tilde{G}_i \prod_{j<i} (1 - \alpha_j \, \tilde{G}_j), \tag{2}$$

where $\tilde{G}_i$ denotes the 2-D projection of Gaussian i. Training proceeds by differentiable rasterisation and minimisation of a combined ℓ_1 and SSIM photometric loss. An *adaptive densification* strategy clones under-reconstructed Gaussians and splits over-large ones, while pruning transparent or redundant primitives.

3DGS achieves real-time rendering (>100 fps at 1080p) with training times of tens of minutes—a dramatic improvement over NeRF—primarily because rasterisation of 2-D Gaussian splats on a GPU tile grid requires no per-pixel ray integration. Its explicit, editable representation makes it attractive for downstream tasks in scene editing, simulation, and embodied AI, where the ability to identify, move, and delete individual scene elements is essential. The primary limitations relative to implicit representations are a *higher memory footprint* (millions of Gaussians per scene) and *limited surface quality* compared to SDF-based methods, both of which have been substantially addressed by the large subsequent literature we survey in Section 5.

2.4 Unified NVS and World Modeling Formulation

We now introduce the formal framework that unifies all methods across the survey. Standard NVS can be written as:

$$\mathcal{M}_\theta : \left(\{\mathbf{I}_i, \boldsymbol{\pi}_i\}_{i=1}^{n}, \; \boldsymbol{\pi}^* \right) \longrightarrow \hat{\mathbf{I}}^*, \quad n \in [1, +\infty). \tag{3}$$

This formulation covers everything from dense multi-view reconstruction (n large, $\mathcal{M}$ an explicit renderer) to single-image generation ($n = 1$, $\mathcal{M}$ a diffusion model) and unconditioned 3-D generation ($n = 0$, $\mathcal{M}$ a generative model with a text or noise input).

The world-modeling extension augments this with a time index t and a sequence of agent actions $\mathbf{a}_{0:t}$:

$$\mathcal{M}_\theta : \left(\{\mathbf{I}_i, \boldsymbol{\pi}_i\}_{i=1}^{n}, \; \mathbf{a}_{0:t}, \, t, \, \boldsymbol{\pi}^* \right) \longrightarrow \hat{\mathbf{I}}_t^*. \tag{4}$$

Eq. (3) is the special case $t = 0$, $\mathbf{a} = \varnothing$. The additional terms expose the three axes of the spectrum introduced in §1: controlling $\boldsymbol{\pi}^*$ corresponds to Axis 1 (input regime); adding the time index t activates Axis 2 (temporal scope); and adding $\mathbf{a}$ activates Axis 3 (controllability).

A complete world model satisfying Eq. (4) must fulfil five desiderata, which we use as a consistent evaluation lens throughout the survey: **D1** geometric consistency across viewpoints; **D2** generalisation from sparse or single-image input; **D3** dynamic scene modelling; **D4** action conditioning; **D5** physical plausibility. No single existing method satisfies all five simultaneously—that gap defines the central open problem and the research frontier we identify in Section 11.

3 Scene Representation Taxonomy

The choice of scene representation is arguably the single most consequential design decision in any NVS system. It determines the fundamental trade-offs a method faces—rendering speed, reconstruction accuracy, memory footprint, generalisability, and editability—and has a direct bearing on whether the representation can serve as a persistent world state for downstream planning or action conditioning. Over the past several years, the field has explored a rich variety of representations, each occupying a different position on the speed–compactness–quality Pareto frontier. We identify four canonical families below and return to each in depth in Sections 4–6.

Implicit representations (MLPs, hash grids) encode the scene as a continuous function $f_\theta(\mathbf{x})$ evaluated at arbitrary 3-D query points. Their key advantages are compactness—a complete scene is stored as a few megabytes of network weights—and continuous differentiability, which supports high-quality reconstruction from dense observations and enables gradients to flow through the rendering process for downstream optimisation. Their central limitation is rendering speed: each pixel requires integrating along a ray through many sequential network evaluations, making real-time rendering impractical without architectural acceleration. NeRF [11] and its successors (Section 4) constitute the canonical family, with subsequent work substituting the MLP for multi-resolution hash grids [156] or tensor decompositions [1] to substantially accelerate training.

Explicit representations (voxels, point clouds, Gaussians) store scene attributes at discrete spatial elements, enabling tile-based or splat-based rasterisation that achieves real-time frame rates through massive GPU parallelism. The trade-off is memory: a faithful 3DGS scene may occupy hundreds of megabytes to gigabytes, and the discrete nature of the representation makes it less amenable to continuous differentiable reasoning. 3D Gaussian Splatting [10] (Section 5) is the dominant current representative, having largely supplanted point-cloud-based methods by incorporating learnable appearance and anisotropic geometry.

Hybrid representations combine an explicit spatial structure—typically a triplane [22] or multi-resolution grid [156]—with a shallow MLP decoder. They strike a practical balance between the quality of implicit models and the speed of explicit ones: the spatial structure provides memory-efficient feature caching, while the MLP decoder retains the capacity for high-frequency appearance modelling. This combination has proven especially powerful for large-scale generalised reconstruction, and it is the architectural backbone of most Large Reconstruction Models (Section 6.2) and EG3D-style 3-D-aware GANs [22].

Compositional representations decompose the scene into semantic entities (objects, backgrounds, agents) with independent sub-representations. This structural inductive bias enables object-level editing and compositional generation [120, 245], but typically requires supervision or segmentation to separate entities. It is also the natural representation for world models that must reason about independently acting objects—vehicles, humans, and manipulable items—each with its own dynamics model. The growing intersection of compositional representations and world modeling is one of the most active research fronts we identify in Section 11.

Each of the four sections that follow (Sections 4–6) develops one branch of this taxonomy in detail. The taxonomy is not exhaustive, and the boundaries between families are blurring: feed-forward Gaussian methods (Section 6) simultaneously exhibit the speed of explicit representations and the generalisability of implicit ones, while SDF-based neural methods (Section 4) occupy an intermediate position between implicit and explicit through their mesh extraction pipeline. We return to the implications of this convergence in Section 8.

4 NeRF: Implicit Neural Radiance Fields

Neural Radiance Fields occupy the foundational stratum of modern NVS. Their introduction by B. Mildenhall and Ng [11] demonstrated, for the first time, that a compact neural parameterisation could represent complex real scenes with view-dependent appearance at a quality that classical geometry-and-texture pipelines could not approach. The subsequent literature has pursued two independent but equally important directions: *accelerating* per-scene NeRF optimisation to make it practical, and *generalising* it beyond per-scene fitting to enable fast inference across diverse, previously unseen scenes. A third, deeply important direction addresses *dynamic and deformable scenes*, which is a necessary prerequisite for world modeling. We survey all three sub-families below, with Table 2 providing a comprehensive cross-family comparison.

4.1 Foundational NeRF Methods

The original NeRF [11] demonstrated that an MLP with positional encoding could faithfully represent complex real scenes with view-dependent appearance from a few dozen posed photographs. Subsequent work systematically addressed its two principal limitations—slow optimisation and poor generalisation—laying the groundwork for the modern NeRF ecosystem. Before examining these extensions, however, we first discuss improvements to the core per-scene optimisation pipeline, as they established the quality and speed baselines against which all later methods are measured.

Accelerating per-scene optimisation. The central bottleneck in vanilla NeRF is the MLP query: every rendered pixel requires hundreds of network evaluations along a sampled ray, making training convergence require hours even on high-end GPUs. The insight that motivated the first generation of accelerated methods was that the large MLP is architecturally overspecified for a fixed scene—much of its capacity is consumed by storing spatial correlations that could be encoded far more efficiently in an explicit spatial data structure.

Instant-NGP [156] realised this insight most completely, replacing the large MLP backbone with a compact multi-resolution *hash grid* that stores feature vectors at a trainable table of spatial positions. The hash encoding is both compact and highly parallelisable on GPU, enabling training convergence in under a minute on a single GPU while matching or exceeding prior quality. TensoRF [1] took a complementary algebraic approach, decomposing the radiance field tensor into low-rank CP and VM components; this tensor factorisation reduces both storage and memory bandwidth without sacrificing the continuous differentiability of the implicit representation. Plenoxels [138] eliminates the MLP entirely, using a sparse 3-D voxel grid with trilinearly interpolated spherical harmonics; the key observation is that view-dependent colour can be represented without a neural network, and the resulting model trains in minutes by avoiding costly backward passes through deep networks. DirectVoxGO [18] accelerates convergence via a two-stage coarse-to-fine voxel grid that aggressively skips empty space during sampling, directly reducing the number of expensive MLP queries to only those near visible surfaces.

Improving visual quality and handling challenging scenes. Even as the acceleration methods were resolving the speed bottleneck, a parallel strand of work addressed systematic quality failures in vanilla NeRF: aliasing at different viewing distances, failure on unbounded scenes, and poor handling of specular and metallic materials.

Mip-NeRF [75] identified aliasing as arising from the point-sampling abstraction of the ray: when a pixel footprint covers a large solid angle—as at large distances or with wide-angle lenses—querying the radiance field at a single point is a poor approximation of the integral over the pixel. Mip-NeRF replaces the point-sampled ray with an *integrated positional encoding* over a conical frustum, substantially reducing aliasing across multiple viewing distances and establishing the first rigorous anti-aliasing framework for neural radiance fields. Mip-NeRF 360 [73] extended this framework to *unbounded* outdoor scenes, which vanilla NeRF and even Mip-NeRF handle poorly because the scene volume is infinite. It resolves this via a non-linear scene parameterisation that contracts the infinite background into a bounded representation, allowing the model to allocate capacity appropriately between the foreground and the distant environment. Zip-NeRF [74] later unified the hash grid of Instant-NGP with the anti-aliasing of Mip-NeRF 360, establishing the current quality frontier for per-scene NeRF by combining the speed of explicit encoding with the rigour of multi-scale anti-aliasing.

NeRF-W [131] addressed a different quality challenge: reconstructing landmarks from uncontrolled internet images, where illumination varies across photographs and transient objects (tourists, cars) are present in some images but not others. It introduces per-image latent appearance codes to absorb illumination variation, together with a transient object model that down-weights inconsistent observations during training. Ref-NeRF [36] tackled the failure of directional encoding to represent specular materials accurately, restructuring the directional encoding into a physically-based reflection model that explicitly predicts diffuse colour, specular tint, and surface roughness. The key contribution is a *reflectance decomposition* that separates the predicted colour into diffuse and specular components, each with its own directional dependence, dramatically improving the rendering of specular and metallic surfaces.

Surface reconstruction. While volume rendering naturally produces accurate alpha composites, it does not enforce a well-defined surface: the density function can be "foggy" near object boundaries, leading to noisy geometry. This matters greatly for applications beyond view synthesis—mesh extraction for rendering in game engines, collision geometry for robotics, or structural analysis in architecture—and motivated a dedicated line of work on SDF-based neural surface reconstruction.

NeuS [122] made the pivotal observation that volume rendering and SDF reconstruction can be unified: by reformulating the volume rendering integral using a logistic density derived from a Signed Distance Function, it obtains watertight, high-quality surface reconstruction without any explicit surface supervision. VolSDF [90] independently derived a rigorous analytical mapping from SDF values to volume density, providing a theoretically grounded alternative and demonstrating that the logistic function used by NeuS is one member of a broader family of valid density functions. BakedSDF [89] takes the SDF-based surface quality one step further by combining high-quality SDF reconstruction with real-time rendering: after optimisation, the learned SDF is extracted into a mesh and the view-dependent appearance is "baked" into a texture, enabling real-time playback of neurally reconstructed scenes.

Table 2: Comprehensive NeRF method comparison. **Category**: PS=per-scene, GEN=generalizable, DYN=dynamic. **Train/Render**: speed relative to vanilla NeRF (Fast/Med/Slow). **Gen.**: generalizes across scenes without per-scene re-training. **Dyn.**: handles dynamic or deformable scenes. **Surf.**: produces explicit surface (mesh/SDF).

Method	Venue	Cat.	Train	Render	Gen.	Dyn.	Surf.	Key Contribution
Per-scene optimisation								
NeRF [11]	ECCV'20	PS	Slow	Slow	–	–	–	MLP + positional encoding
NeRF-W [131]	CVPR'21	PS	Slow	Slow	–	–	–	In-the-wild appearance codes
Mip-NeRF [75]	ICCV'21	PS	Slow	Slow	–	–	–	Integrated positional encoding
NeuS [122]	NeurIPS'21	PS	Slow	Slow	–	–	✓	SDF-based volume rendering
VolSDF [90]	NeurIPS'21	PS	Slow	Slow	–	–	✓	Analytical SDF-to-density
Plenoxels [138]	CVPR'22	PS	Fast	Fast	–	–	–	No-MLP voxel grid + SH
TensoRF [1]	ECCV'22	PS	Med	Fast	–	–	–	CP/VM tensor decomposition
DirectVoxGO [18]	CVPR'22	PS	Fast	Fast	–	–	–	Two-stage coarse-to-fine voxel
Instant-NGP [156]	TOG'22	PS	Fast	Fast	–	–	–	Multi-resolution hash grid
Mip-NeRF360 [73]	CVPR'22	PS	Slow	Slow	–	–	–	Unbounded scene contraction
Ref-NeRF [36]	CVPR'22	PS	Slow	Slow	–	–	–	Physics-based reflection model
Zip-NeRF [74]	ICCV'23	PS	Fast	Fast	–	–	–	Hash grid + anti-aliasing
BakedSDF [89]	SIGGRAPH'23	PS	Med	Fast	–	–	✓	SDF mesh extraction + baking
Generalizable methods								
pixelNeRF [7]	CVPR'21	GEN	Fast	Slow	✓	–	–	Pixel-aligned CNN features
IBRNet [127]	CVPR'21	GEN	Fast	Slow	✓	–	–	Cross-view transformer aggregation
MVSNeRF [2]	ICCV'21	GEN	Fast	Slow	✓	–	–	Cost-volume geometry prior
SRT [117]	CVPR'22	GEN	Fast	Fast	✓	–	–	Set-latent scene representation
GNT [143]	arXiv'22	GEN	Fast	Med	✓	–	–	Fully attention-based, no NeRF MLP
NerfDiff [48]	ICML'23	GEN	Fast	Slow	✓	–	–	Diffusion prior for single-view
ZeroRF [134]	arXiv'23	GEN	Med	Slow	∼	–	–	Zero-pretrain sparse-view 360°
FrameNeRF [88]	arXiv'24	GEN	Fast	Slow	✓	–	–	Video frames as pseudo views
Dynamic & deformable methods								
D-NeRF [6]	CVPR'21	DYN	Slow	Slow	–	✓	–	Canonical deformation network
NSFF [218]	CVPR'21	DYN	Slow	Slow	–	✓	–	Explicit scene flow field
Nerfies [78]	ICCV'21	DYN	Slow	Slow	–	✓	–	Per-frame latent deformation
HyperNeRF [79]	TOG'21	DYN	Slow	Slow	–	✓	–	Higher-dimensional ambient space
Non-Rigid NeRF [42]	ICCV'21	DYN	Slow	Slow	–	✓	–	Monocular ray bending
DyNeRF [219]	CVPR'22	DYN	Slow	Slow	–	✓	–	Time-conditioned multi-view NeRF
TiNeuVox [64]	SIGGRAPH'22	DYN	Med	Med	–	✓	–	Voxel-accelerated dynamic NeRF
K-Planes [137]	CVPR'23	DYN	Fast	Fast	–	✓	–	4-D plane decomposition
HexPlane [21]	CVPR'23	DYN	Fast	Fast	–	✓	–	Concurrent 4-D plane factorisation
NeRFPlayer [151]	TOG'23	DYN	Med	Med	–	✓	–	Static/deforming/new region decomp.
Soup-of-Planes [40]	ICCV'23	DYN	Fast	Fast	–	✓	–	Planar fitting for casual video
Pseudo-GDV [240]	CVPR'24	DYN	Med	Slow	–	✓	–	Static + dynamic inpainting

Together, these three sub-families—acceleration, quality, and surface—resolved the core technical limitations of vanilla NeRF and set the stage for the generalisation and dynamics work discussed next.

Table 2 provides a comprehensive comparison of NeRF variants across all three sub-families: per-scene optimisation, generalizable, and dynamic methods.

4.2 Generalizable NeRF

Per-scene NeRF achieves high quality but at the cost of lengthy per-scene optimisation, which is fundamentally incompatible with applications requiring rapid scene understanding—robotic manipulation, autonomous navigation, and real-time AR all require scene understanding on the order of seconds, not hours. *Generalisable* methods address this bottleneck by learning a shared prior across many training scenes, enabling fast inference on previously unseen scenes without any per-scene optimisation. The key design challenge is determining how to encode the reference views into a scene representation that the NeRF decoder can query efficiently.

pixelNeRF [7] established the foundational paradigm: it conditions the NeRF MLP on pixel-aligned CNN features extracted from reference views, so that at any query point along a ray, the model can attend to the corresponding image-space features. This pixel-aligned conditioning enables single-image reconstruction but inherits the NeRF MLP's slow rendering speed. IBRNet [127] improved upon this by using a transformer to aggregate features from multiple reference views at query points along the ray, learning to weight each reference view by its relevance to the current query location and thus producing smoother, more consistent geometry. MVSNeRF [2] introduced explicit geometric reasoning by building a cost volume from stereo correspondences, providing a geometry-aware feature representation that substantially improves reconstruction under large view baselines.

A particularly important direction is the elimination of the NeRF MLP altogether. GNT [143] dispenses with the volumetric rendering MLP, replacing it with a two-stage transformer: the first stage encodes reference views into a view-independent scene representation, and the second renders the target view via cross-attention. This architecture learns geometric reasoning as an emergent property of attention rather than an explicit inductive bias, achieving better generalisation on scenes that violate the assumptions of cost-volume methods. SRT [117] pursues a similar philosophy with a set-latent architecture, encoding an arbitrary number of views into a fixed-size scene representation without pose assumptions, enabling pose-free inference that anticipates the DUSt3R line of work.

More recent approaches have augmented generalisable NeRF with generative priors to handle the under-constrained regime of very sparse input. NerfDiff [48] incorporates a diffusion model to synthesise additional virtual views that supplement the sparse reference set, effectively using the diffusion prior as a learned geometric hallucinator. ZeroRF [134] achieves competitive sparse-view 360° reconstruction via per-scene optimisation with a factorised feature grid and no pretraining, demonstrating that careful regularisation can substitute for generalisation in some sparse-input regimes.

Few-shot and test-time methods. The sparse-input regime has attracted sustained attention because it is the most practically relevant: in deployment, reference images are scarce. FrameNeRF [88] exploits the temporal redundancy in video by treating consecutive frames as dense pseudo-views, dramatically increasing effective input density from monocular video without requiring multi-camera hardware. Recollection from Pensieve [246] takes a step further, learning NVS from uncalibrated internet videos without any posed supervision, treating pose estimation as an internal latent variable. Test-Time Completion [14] repurposes natural video completion as a zero-shot NVS prior at test time, demonstrating that large video models already encode sufficient spatial knowledge for competitive NVS without additional training. Mixed Neural Voxels [172] accelerates multi-view video synthesis by decomposing scenes into static and dynamic voxel grids, exploiting the temporal smoothness of most dynamic scenes.

Taken together, generalisable NeRF methods represent a conceptual shift from the per-scene paradigm toward the *amortised inference* view of neural rendering: a single trained model encodes a universal geometric prior that can be applied to any new scene at inference time. This shift is a prerequisite for world modeling—a world model that required hours of optimisation for each new scene would be useless in real-time decision-making settings—and it directly motivates the feed-forward reconstruction models discussed in Section 6.

4.3 Dynamic NeRF

Modelling non-rigid dynamic scenes is a central challenge for world modeling, as it requires the representation to capture not just spatial geometry but also the temporal evolution of scene structure. The fundamental difficulty is that a dynamic scene lacks a fixed canonical frame: the geometry changes continuously, and a naive extension of static NeRF to incorporate time as an additional input dimension often fails to model large deformations faithfully.

The first generation of dynamic NeRF methods addressed this through the *canonical deformation* paradigm. D-NeRF [6] introduces a deformation network that maps each spacetime query point (x, t) to a corresponding location in a canonical static space, where a standard NeRF computes appearance; at test time, novel views at any timestamp are rendered by composing the deformation and canonical fields. Nerfies [78] refines this idea for casual monocular video of human subjects, where the long-range correlations of body motion make pure deformation network predictions unreliable; it instead uses per-frame latent deformation codes with regularisation to prevent physically implausible deformations. HyperNeRF [79] generalises further to scenes with topological changes—mouth opening, paper tearing—which require changes in surface connectivity that smooth deformations cannot represent. It embeds the canonical space in a higher-dimensional ambient space, allowing the "slice" seen at each timestep to represent scenes of changing topology.

Alongside the deformation paradigm, Neural Scene Flow Fields (NSFF) [218] proposed an orthogonal approach: explicitly modelling the 3-D velocity field of the scene, enabling supervision from monocular depth estimation and optical flow and providing an intuitive physical interpretation of motion. Non-Rigid NeRF [42] reconstructs and synthesises novel views of dynamic scenes from monocular video with explicit ray bending, demonstrating that accurate reconstruction from a single moving camera is achievable even for complex non-rigid motion. DyNeRF [219] extends NeRF to multi-view video by conditioning on time, providing a cleaner experimental setting with strong geometric grounding from simultaneous cameras.

The second generation of dynamic NeRF methods focused on efficiency, motivated by the observation that per-scene dynamic NeRF optimisation is even slower than its static counterpart. TiNeuVox [64] accelerates dynamic NeRF by replacing the deformation-field MLP with a time-conditioned voxel feature grid, reducing training time from days to hours. The key architectural innovation came with K-Planes [137] and HexPlane [21], which independently proposed the same elegant solution: decompose 4-D space-time into six feature planes—three spatial (XY, XZ, YZ) and three spatial-temporal (XT, YT, ZT)—each encoded as a learned 2-D grid, making the architecture highly memory-efficient and interpretable. Their simultaneous publication demonstrated the generality of the plane-based 4-D decomposition idea and established it as the dominant acceleration paradigm for dynamic NeRF. NeRFPlayer [151] took a complementary approach to efficiency by separately encoding static, deforming, and newly-appearing scene regions, allowing the model to allocate capacity according to local motion statistics.

More recent work has pushed toward casual, monocular dynamic video synthesis. Soup-of-Planes [40] enables fast view synthesis of casual monocular videos by fitting a lightweight set of planar scene representations; this simple parameterisation is surprisingly effective for the forward-facing camera motions common in casual video. Pseudo-Generalised Dynamic View [240] synthesises dynamic novel views from monocular video by combining static reconstruction with per-frame dynamic inpainting, separating the estimation problem into tractable sub-problems. Decoupled Dynamic Video [177] disentangles foreground dynamics and background statics, assigning different model capacities to each and enabling cleaner decomposition of casual monocular videos.

The trajectory of dynamic NeRF research mirrors that of static NeRF: from proof-of-concept demonstrations requiring dense camera rigs to fast, practical methods operating on casual monocular video. This trajectory is directly connected to the world-modeling agenda: a world model must reason about dynamic scenes from single-camera observations, precisely the regime where the most recent dynamic NeRF methods operate. The dynamic NeRF methods are included in the comprehensive Table 2 above.

5 3D Gaussian Splatting

Despite the quality advances of the NeRF family, a fundamental bottleneck remained: volumetric ray marching requires hundreds of sequential MLP evaluations per ray, making real-time rendering on commodity hardware essentially unachievable. 3D Gaussian Splatting (3DGS) [10] resolved this constraint decisively, achieving over 100 fps at 1080p resolution by replacing the continuous radiance field with a collection of explicit, anisotropic Gaussian primitives that are rendered via tile-based rasterisation. Beyond raw speed, the representation is fully explicit and directly editable: individual Gaussians can be identified, moved, duplicated, and removed without re-optimisation. These properties have opened a new chapter in NVS research, generating a large and rapidly expanding literature that we organise here into four thematic directions: quality and anti-aliasing improvements, compression and memory efficiency, surface reconstruction, and scene editing and compositionality. Table 3 provides a comprehensive cross-family comparison.

5.1 Foundational 3DGS

3DGS [10] achieved real-time ($>$100 fps at 1080p) NVS with training times of 30–45 minutes on commodity hardware, surpassing NeRF-quality baselines on standard benchmarks. Its three key innovations are: (i) anisotropic Gaussian primitives that align naturally with planar scene structures; (ii) tile-based rasterisation that enables massively parallel GPU rendering; (iii) adaptive densification that automatically allocates capacity to under-reconstructed regions. The resulting representation is fully explicit and editable, opening new directions in scene editing, simulation, and avatar rendering.

Quality and anti-aliasing. With the rendering speed bottleneck resolved, the focus shifted immediately to improving the visual quality of Gaussian-based rendering, particularly in regimes that exposed the approximations made by the original method.

Mip-Splatting [227] identifies aliasing in vanilla 3DGS as arising from the approximation of 3-D Gaussian convolution during 2-D projection. It introduces a 3-D smoothing filter derived from the sampling footprint, enabling accurate multi-scale rendering and substantially reducing floaters at large distances. This is the direct Gaussian-space analogue of the integrated positional encoding insight from Mip-NeRF, and it similarly establishes the anti-aliasing baseline for the 3DGS literature. AbsGS [226] revisits the densification criterion: replacing the mean gradient with an absolute-value measure improves coverage in thin structures and reduces needle-like artifacts near object boundaries, demonstrating that the adaptive densification heuristic of vanilla 3DGS leaves systematic quality gaps that a principled gradient measure can address.

Compression and memory efficiency. A single 3DGS scene may require 300 MB–1 GB of storage, limiting deployment on mobile or embedded platforms and making it impractical to maintain multiple scenes in GPU memory simultaneously. Addressing this bottleneck has motivated a dedicated compression sub-field that seeks to reduce storage cost while preserving rendering quality.

Scaffold-GS [113] introduces a sparse set of anchor primitives from which neural networks predict neighbouring Gaussian attributes; this structured representation reduces storage by 3–5$\times$ with comparable quality. Rather than compressing an unstructured point cloud post-hoc, it imposes architectural structure during optimisation, ensuring that each Gaussian is efficiently described relative to its spatial context. LightGaussian [217] achieves 15$\times$ compression via global Gaussian pruning based on visibility contribution and subsequent vector quantisation of attributes, demonstrating that the vast majority of Gaussians contribute negligibly to the final render and can be aggressively pruned without perceptible quality loss. EAGLES [139] applies residual vector quantisation for fine-grained attribute compression, pushing the quantisation of Gaussian attributes beyond scalar to multi-level residual codebooks. HAC [199] proposes a hash-grid-based context model that predicts per-Gaussian attribute entropy, enabling learned arithmetic coding for extreme compression ratios by exploiting spatial correlations between neighbouring Gaussians that simpler quantisation methods ignore.

Surface reconstruction. The Gaussian primitive is an inherently volumetric object: its density falls off smoothly in all directions with no well-defined surface. This makes mesh extraction from 3DGS non-trivial, yet many downstream applications—physical simulation, collision checking, UV texture mapping, and export to standard graphics pipelines—require explicit surface meshes. A dedicated line of work has addressed this by encouraging Gaussians to adopt disc-like or planar configurations aligned with scene surfaces.

SuGaR [49] aligns Gaussians to the scene surface by adding a regularisation term that encourages Gaussians to become flat discs at the surface, followed by mesh extraction and joint Gaussian–mesh refinement. This two-stage approach leverages the speed of Gaussian optimisation for initial scene reconstruction and the geometric precision of mesh refinement for final surface quality. 2DGS [63] takes a more principled approach by replacing 3-D Gaussians with 2-D disc primitives, which by construction lie on surfaces and produce accurate normal maps; the representation guarantees surface alignment without post-hoc regularisation. Gaussian Opacity Fields (GOF) [228] fuses Gaussian Splatting with a continuous opacity field for topologically correct surface extraction in unbounded scenes, addressing the failure mode of earlier methods on complex open-world geometry.

Scene editing and compositionality. The explicit Gaussian representation is particularly amenable to editing: individual primitives can be identified, moved, duplicated, or removed. This editability has inspired a line of work on semantically-aware and compositional Gaussian scenes that bridges NVS and scene understanding.

GaussianEditor [200] harnesses a large vision-language model to select semantically relevant Gaussians from a text description and optimises their appearance with a diffusion prior. The approach demonstrates that the VLM's semantic segmentation can be transferred to the Gaussian domain via rendered-view supervision, enabling language-guided 3-D editing without any 3-D annotation. Gaussian Grouping [118] augments each Gaussian with a learned identity field, trained with SAM [83] masks from rendered views, enabling instance-level part decomposition and editing. By propagating 2-D instance masks into 3-D Gaussian space via the differentiable renderer, it achieves zero-shot

Table 3: Comprehensive 3D Gaussian Splatting method comparison. **Category**: STATIC=per-scene static, DYN=dynamic/deformable, FF=feed-forward generalizable. **RT**: real-time rendering (>30 fps). **Comp.**: dedicated compression or storage reduction. **Surf.**: explicit surface extraction. **Unp.**: works on unposed images. **Input**: minimum input views for FF methods.

Method	Venue	Cat.	RT	Comp.	Surf.	Edit.	Unp.	Input	Key Contribution
Static per-scene									
3DGS [10]	TOG'23	S	✓	–	–	–	–	MV	Tile-based Gaussian splatting
Mip-Splatting [227]	CVPR'24	S	✓	–	–	–	–	MV	3-D smoothing filter, anti-alias
AbsGS [226]	NeurIPS'24	S	✓	–	–	–	–	MV	Absolute-gradient densification
Scaffold-GS [113]	CVPR'24	S	✓	✓	–	–	–	MV	Anchor-driven neural attributes
LightGaussian [217]	NeurIPS'23	S	✓	✓	–	–	–	MV	Visibility pruning + VQ, 15×
Compact3D [92]	arXiv'24	S	✓	✓	–	–	–	MV	k-means codebook compression
EAGLES [139]	ECCV'24	S	✓	✓	–	–	–	MV	Residual vector quantisation
HAC [199]	ECCV'24	S	✓	✓	–	–	–	MV	Hash context entropy coding
SuGaR [49]	CVPR'24	S	✓	–	✓	∼	–	MV	Flat-disc + mesh joint opt.
2DGS [63]	SIGGRAPH'24	S	✓	–	✓	–	–	MV	2-D disc primitives
GOF [228]	arXiv'24	S	✓	–	✓	–	–	MV	Opacity field for surface
GaussianEditor [200]	CVPR'24	S	✓	–	–	✓	–	MV	LLM-guided semantic editing
Gaussian Grouping [118]	ECCV'24	S	✓	–	–	✓	–	MV	SAM-based identity field
GALA3D [245]	ICML'24	S	✓	–	–	✓	–	MV	LLM-layout compositional gen.
LERF [68]	ICCV'23	S	–	–	–	∼	–	MV	CLIP language grounding in NeRF
LangSplat [128]	CVPR'24	S	✓	–	–	∼	–	MV	Language field in 3DGS
Dynamic & deformable									
4DGS [44]	CVPR'24	D	✓	–	–	–	–	MV	4-D Gaussian + temporal field
Deformable3DGS [224]	ICLR'24	D	✓	–	–	–	–	Mono	Canonical deformation network
SpacetimeGaussians [225]	CVPR'24	D	✓	–	–	–	–	MV	Polynomial time trajectories
SC-GS [201]	CVPR'24	D	✓	–	–	–	–	Mono	Sparse control point deform.
Gaussian-Flow [107]	CVPR'24	D	✓	–	–	–	–	MV	Fourier polynomial motion
Shape-of-Motion [173]	arXiv'24	D	–	–	–	–	–	Mono	Per-Gaussian 3-D motion
Dynamic 3DGS [114]	3DV'24	D	✓	–	–	–	–	MV	Persistent Gaussian tracking
GauFRe [106]	arXiv'24	D	✓	–	–	–	–	Mono	Flow-regularised deformation
MoVieS [154]	arXiv'24	D	✓	–	–	–	–	Mono	4-D synthesis in ∼1 second
SC-4D [82]	arXiv'24	D	–	–	–	–	–	Mono	Self-calibrating 4-D from video
Reconstruct-Inpaint [55]	arXiv'24	D	–	–	–	–	–	Mono	Recon + inpaint + TTF pipeline
Articulated-GS [103]	NeurIPS'24	D	✓	–	–	–	–	MV	Template-free articulated 3DGS
EVA-Gaussian [62]	arXiv'24	D	✓	–	–	–	–	MV	Real-time human NVS
Feed-forward generalizable									
pixelSplat [35]	CVPR'24	FF	✓	–	–	–	–	2	Stereo epipolar features
MVSplat [32]	ECCV'24	FF	✓	–	–	–	–	2–6	Cost-volume geometry prior
TranSplat [234]	arXiv'24	FF	✓	–	–	–	–	2+	Transformer MV prediction
FSGS [231]	CVPR'24	FF	✓	–	–	–	–	3	Gaussian unpooling, few-shot
DepthSplat [195]	arXiv'24	FF	✓	–	–	–	–	2+	Joint depth + Gaussian pred.
Splatt3R [150]	arXiv'24	FF	✓	–	–	–	✓	2	DUSt3R backbone, unposed
TriplaneGaussian [222]	CVPR'24	FF	✓	–	–	–	–	1	Triplane + Gaussian, single-view

part decomposition that requires no 3-D supervision. LERF [68] and LangSplat [128] both ground language into 3-D neural representations, with LERF doing so in NeRF space and LangSplat directly in the Gaussian representation; together they demonstrate that open-vocabulary 3-D querying—"show me all chairs in this scene"—is achievable with language-image pretraining. GALA3D [245] uses an LLM to generate a 3-D layout from text and initialises Gaussians per-object, enabling compositional scene generation that extends the scene-level 3DGS capabilities beyond reconstruction to text-driven synthesis of entirely novel multi-object scenes.

The synthesis of editing-focused methods reveals an important pattern: the Gaussian representation is increasingly serving as the interface between 2-D foundation models (SAM, CLIP, LLMs) and 3-D scene understanding, because the differentiable rasteriser provides a natural gradient bridge from rendered-view 2-D supervision to 3-D primitive attributes. This architectural pattern—2-D supervision → Gaussian attributes—is likely to become even more central as 2-D foundation models continue to improve.

Table 3 provides a comprehensive view of the 3DGS family.

5.2 Dynamic 3DGS

Static 3DGS, while powerful, is limited to scenes that remain fixed over time. Extending Gaussian splatting to dynamic scenes requires mechanisms to model the *motion* of Gaussian primitives over time—a challenge that has attracted considerable attention given the representation's unique combination of explicit, editable structure and real-time rendering speed. The key design question is whether to model motion as a *deformation field* applied to canonical Gaussians, or as an *intrinsic temporal attribute* directly parameterising each Gaussian's trajectory.

4DGS [44] takes the second approach, extending Gaussians into 4-D space-time by pairing each primitive with a temporal radiance field that modulates its position and appearance as a function of time; this formulation enables real-time rendering of complex dynamic scenes such as sports and dance. SpacetimeGaussians [225] assigns polynomial time trajectories directly to each Gaussian's position and rotation, providing a compact and analytically differentiable motion representation that avoids the computational cost of an implicit deformation network. Deformable3DGS [224] and SC-GS [201] adopt the canonical deformation approach, following the D-NeRF paradigm but in the Gaussian domain: Deformable3DGS applies a global deformation network to all Gaussians, while SC-GS uses sparse control points that provide a physically interpretable, structure-aware deformation field particularly suited for articulated and human body scenes.

Gaussian-Flow [107] models Gaussian dynamics via polynomial motion fields in Fourier space, enabling compact temporal parameterisation that naturally captures periodic motions such as walking and repetitive industrial machinery. Shape-of-Motion [173] takes a more ambitious approach, jointly estimating per-Gaussian 3-D motion and global scene flow from monocular video, operating in the most challenging input regime. Dynamic 3D Gaussians [114] tracks scene dynamics through persistent Gaussian identities across views, treating the tracking and NVS problems as jointly constrained by the photometric consistency of the Gaussian primitives.

GauFRe [106] and MoVieS [154] both target the *efficiency* end of the dynamic 3DGS spectrum: GauFRe regularises the deformation field with dense optical flow to produce temporally coherent results from monocular video, while MoVieS generates motion-aware 4-D dynamic views in approximately one second via an efficient Gaussian motion estimation pipeline. Self-Calibrating 4D [82] addresses the practical challenge that monocular dynamic videos lack calibrated camera parameters, jointly estimating camera intrinsics and extrinsics alongside the Gaussian deformation. Reconstruct-Inpaint-Finetune [55] takes a pragmatic three-stage approach—first reconstructing the static scene, then inpainting dynamic regions with a generative model, and finally refining the result at test time—that produces high-quality dynamic novel views without an end-to-end dynamic 3DGS model. EVA-Gaussian [62] achieves real-time human NVS under diverse multi-view settings. Articulated Gaussian [103] enables template-free articulated Gaussian Splatting for real-time reposable dynamic view synthesis.

Surveying the dynamic 3DGS landscape reveals a common tension: deformation-field methods provide high quality but require per-scene optimisation, while trajectory-attribute methods are faster but less expressive for complex non-rigid motion. The monocular dynamic synthesis problem remains substantially harder than the multi-view case due to the depth-motion ambiguity, and the best current methods still rely on additional supervision from optical flow and monocular depth estimates to constrain the otherwise ill-posed optimisation.

5.3 Feed-Forward Generalizable 3DGS

The feed-forward 3DGS paradigm inherits the architectural pattern established by generalisable NeRF—amortise scene-specific computation into a pretrained model—while retaining the rendering speed advantage of the Gaussian representation. The result is a family of models that synthesise novel views in a single forward pass, without any per-scene optimisation, at real-time rendering speed. This combination of generalisation and rendering efficiency is precisely what downstream applications in robotics and autonomous driving require.

pixelSplat [35] established the feed-forward 3DGS paradigm by predicting Gaussian primitives from stereo image pairs in a single forward pass, using epipolar attention to exploit the known geometric relationship between the two cameras. The model implicitly solves stereo matching, depth estimation, and Gaussian attribute prediction in a unified architecture. MVSplat [32] adds explicit cost-volume priors for better geometry under challenging viewpoints, demonstrating that the multi-view stereo geometry prior—which proved essential in the generalisable NeRF literature—is equally beneficial in the Gaussian domain. TranSplat [234] introduces a transformer architecture for multi-view Gaussian prediction that attends across all input views jointly, enabling better utilisation of cross-view geometric consistency cues.

Splatt3R [150] takes a qualitatively different approach by building on DUSt3R [174], which predicts dense 3-D point maps directly from unposed image pairs. By treating the DUSt3R pointmaps as Gaussian positions and learning only the Gaussian attributes (colour, opacity, scale) via a lightweight decoder, Splatt3R achieves unposed image reconstruction—no camera calibration required—which dramatically simplifies the capture pipeline. FSGS [231] enables few-shot Gaussian Splatting via Gaussian unpooling and monocular depth supervision, specifically addressing the case where only three or fewer reference views are available. DepthSplat [195] jointly predicts depth and Gaussians in a single model, sharing representations between the depth estimation and Gaussian prediction tasks to improve geometry quality. TriplaneGaussian [222] combines triplane and Gaussian representations in a transformer-based single-view reconstruction model, using the triplane as an intermediate 3-D feature representation that is then decoded into Gaussian primitives.

The feed-forward generalizable methods are included in the comprehensive Table 3 above (FF category).

6 Generalisable Feed-Forward 3D Reconstruction

The per-scene paradigm that dominates the NeRF and classical 3DGS literature carries a fundamental limitation: every new scene requires a fresh optimisation loop lasting tens of minutes to hours. This is incompatible with real-world applications where new scenes are encountered continuously—autonomous vehicles observing novel road segments, robots entering unfamiliar rooms, AR systems overlaying content on new environments. The *feed-forward reconstruction* paradigm directly addresses this bottleneck: a single, amortised model trained on large datasets of 3-D assets or multi-view captures learns to predict a complete 3-D scene representation from one or a few images in a single forward pass.

This section covers two complementary approaches to this goal. We first discuss the pose-free foundation model line of work (DUSt3R, MASt3R, and successors), which treats reconstruction as a direct regression problem from image pairs to point maps without requiring camera calibration. We then examine the Large Reconstruction Model (LRM) paradigm, which uses large transformer models pretrained on 3-D asset repositories to predict NeRF or Gaussian scenes from a single image. Together, these two families represent the current state-of-the-art in generalised single-image and sparse-view 3-D understanding.

6.1 Pose-Free and Foundation Models

The classical 3-D reconstruction pipeline requires camera calibration as a prerequisite: Structure-from-Motion [145] must be run to estimate camera poses before any NeRF or 3DGS optimisation can begin. This prerequisite is a significant practical obstacle—internet-sourced images rarely come with calibration information, and running SfM on every image collection adds latency. A new generation of *pose-free* models eliminates this prerequisite by learning to predict 3-D geometry directly from image pairs, treating camera pose estimation as an internal computation rather than an external preprocessing step.

DUSt3R [174] is the landmark work of this direction. Rather than predicting depth or disparity (which require calibration to convert to metric 3-D), DUSt3R predicts *dense point maps*: for each pixel in both input images, it directly regresses the 3-D coordinate in a shared Euclidean frame. The key insight is that a ViT backbone with cross-image attention can jointly reason about both images and implicitly solve the correspondence and geometry problems in a single forward pass. For collections of more than two images, DUSt3R aligns pairwise predictions via a global optimisation that finds the rotation and translation that minimises pairwise reprojection error, producing a metrically accurate 3-D reconstruction without any explicit SfM.

MASt3R [93] extends DUSt3R by adding a matching head that produces dense feature descriptors alongside the point map predictions. This addition enables accurate metric reconstruction from internet-sourced photo pairs where appearance varies substantially—a regime where vanilla DUSt3R's purely geometric predictions can accumulate errors—and directly enables downstream applications such as localisation and structure-aware image retrieval.

MonST3R [236] extends DUSt3R to dynamic video sequences, where the scene contains independently moving objects. By jointly estimating camera motion and scene-object motion within the same point-map regression framework, MonST3R handles the dynamic case that DUSt3R implicitly assumes away, substantially broadening the practical applicability of pose-free reconstruction to casually captured video.

VGGT [169] represents the most recent advance in this line, proposing a Visual Geometry Grounded Transformer that predicts a comprehensive set of geometric quantities—point maps, depth, camera parameters, and feature descriptors—from an arbitrary number of views in a single transformer forward pass. Its key innovation is the integration of structured geometric tokens that provide an explicit 3-D coordinate system for the transformer's attention operations, enabling the model to reason jointly about all geometric quantities rather than treating each as an independent prediction head.

The pose-free foundation models collectively demonstrate that large-scale pretraining on diverse 3-D data can substitute for the carefully designed geometric pipelines that classical reconstruction required, opening the door to reconstruction from casually captured, unposed image collections. This generalisation is a direct enabler of world modeling at scale, where collecting calibrated sensor data for every new environment is impractical.

6.2 Large Reconstruction Models (LRM)

While the pose-free models address multi-view reconstruction, a parallel line of work targets an even more challenging regime: reconstructing a complete 3-D scene from a *single* image. Single-image reconstruction is fundamentally under-constrained without a strong prior—the imaging process collapses 3-D geometry to a 2-D projection, discarding all depth information and occluded geometry—and requires a model that has internalised the distribution of plausible 3-D shapes for the objects and scenes it encounters. The Large Reconstruction Model (LRM) paradigm [202] provides this prior through large-scale pretraining: a transformer trained on hundreds of thousands of 3-D assets from Objaverse [37] learns to map a single input image to a complete triplane representation of a NeRF, achieving complete 3-D reconstruction in under five seconds.

The LRM [202] architecture uses a DINO-ViT encoder followed by a cross-attention transformer that maps image tokens to triplane tokens, which are decoded by a NeRF MLP. The key insight is that the transformer can amortise the geometric reasoning of per-scene NeRF optimisation into its weights via large-scale pretraining: by training on hundreds of thousands of objects with known ground-truth geometry, the model learns to *recognise* plausible 3-D shapes rather than *optimise* them. Instant3D [96] decouples appearance generation from geometry prediction, first generating sparse multi-view images with a diffusion model and then feeding the result to an LRM-style reconstructor. This two-stage design allows each component to be trained on different data and to be improved independently, and it achieves text-to-3D generation in under five seconds—a remarkable speed relative to the hours required by SDS-based alternatives. OpenLRM [59] reproduces the LRM pipeline in an open-source setting, enabling broader research access to the paradigm.

The Gaussian-based LRM branch has proven especially popular for its additional rendering speed advantage. LGM [160] generates multi-view images and lifts them to a 3DGS scene via an asymmetric U-Net, achieving 3-D generation in seconds at commercial-grade quality while retaining the real-time rendering advantage of the Gaussian representation. GRM [196] predicts 3D Gaussians directly from image features using a sparse-view transformer, operating entirely at the pixel level without any intermediate NeRF or triplane representation; this pixel-level prediction enables finer spatial resolution in the output Gaussian set. Gamba [58] addresses the computational bottleneck of the transformer's quadratic attention cost by replacing it with a state-space model (Mamba) for $\mathcal{O}(N)$ sequence length scaling, making long-context 3-D inference from high-resolution inputs feasible. GS-LRM [235] combines a transformer backbone with a 3DGS output head, training on multi-view sequences from Objaverse; it achieves state-of-the-art single-image reconstruction quality by jointly training the visual encoder and the Gaussian decoder end-to-end on reconstruction quality.

Two non-parametric reconstruction models serve as particularly strong baselines in the feed-forward regime and additionally underpin the next generation of generalised Gaussian reconstructors. DUSt3R [174] and MASt3R [93]—already discussed in the pose-free section—provide unposed pairwise point maps that Splatt3R [150] then converts to Gaussian splatting scenes via a lightweight attribute prediction head, combining the geometric reliability of DUSt3R with the rendering speed and editability of 3DGS.

Implications for World Modeling. LRM-style models demonstrate that a single transformer, with sufficient data and compute, can learn a universal spatial prior—the same capacity required by world models for instant scene initialisation from single-frame observations. This is directly relevant to the world-modeling agenda: a world model that must initialise a 3-D scene representation from a single observation (a camera frame at the start of a robot episode, or a single photo of a room to be manipulated) can leverage the LRM prior to obtain a geometrically coherent starting point, rather than requiring dense multi-view capture. We return to this connection in Section 8.

6.3 Multi-View Stereo in the Neural Era

Classical MVS has been reinvigorated by deep learning. MVSNet [206] introduced cost-volume-based deep MVS, using a differentiable homography warping to build a cost volume from multiple reference views and predicting per-pixel depth via a 3-D CNN. IterMVS [168] and Uni-MVSNet [124] improved accuracy and generalisation through iterative GRU-based refinement and unified feature matching, respectively. These methods provide the geometry backbone for downstream NVS: their cost-volume feature matching is directly incorporated into generalisable NeRF (MVSNeRF [2]) and feed-forward Gaussian methods (MVSplat [32]), establishing a productive synergy between the classical MVS and neural NVS communities that continues to be productive.

7 Generative Paradigms for Novel View Synthesis

The emergence of deep generative models—first GANs and then diffusion models—has fundamentally altered the landscape of NVS. Where reconstruction-based methods infer a scene representation from observations and render it deterministically, *generative* methods learn a distribution over plausible visual completions given partial observations. This distinction is not merely architectural: it represents a shift from *interpolation* (rendering within the observed convex hull of viewpoints) to *extrapolation* (hallucinating plausible content beyond it). Generative methods are therefore the primary vehicle for the one-image-to-3D, text-to-3D, and single-image world-generation capabilities that bring NVS into the world-modeling regime.

We organise the generative NVS landscape into four paradigms, ordered chronologically and by increasing capability. First, 3-D-aware GANs (§7.1) embed radiance fields inside GAN generators, enabling implicit 3-D supervision from unposed 2-D image collections. Second, pose-conditioned and camera-trajectory diffusion models (§7.2) use the pre-trained image and video priors of large-scale diffusion models for single-image and multi-view NVS with explicit camera conditioning. Third, Score Distillation Sampling (§7.3) repurposes 2-D diffusion models as differentiable score functions for optimising explicit 3-D representations. Fourth, dedicated multi-view diffusion architectures (§7.4) design models from scratch for 3-D-consistent generation at scale. Table 4 at the end of this section provides a structured comparison across all four paradigms.

7.1 3-D-Aware GANs: Learning Geometry from 2-D Supervision

The first wave of generative NVS methods embedded radiance fields inside GAN generators, enabling implicit 3-D supervision from unposed 2-D image collections—a paradigm that proved influential precisely because it required no 3-D ground truth. The central insight is that if the generator is constrained to produce images by rendering a 3-D representation, the adversarial training signal is sufficient to recover approximate 3-D geometry from 2-D supervision alone.

GRAF [80] pioneered this direction by conditioning a NeRF generator on a latent shape-appearance code, training with a 2-D discriminator. While GRAF established the GAN-NeRF framework, it was limited to low resolution and trained slowly due to the computational cost of volumetric rendering. pi-GAN [41] improved the NeRF backbone with periodic activations (SIREN [164]), achieving substantially better multi-view consistency and demonstrating that the choice of activation function significantly impacts the sharpness and coherence of the generated geometry. GIRAFFE [120] introduced scene compositionality by combining foreground and background neural feature fields, enabling object-level editing without explicit segmentation—a valuable capability for world modeling that requires reasoning about independently acting scene entities.

EG3D [22] is the landmark method of this era. Its *triplane* representation—three axis-aligned 2-D feature maps decoded by a shallow MLP—reduces rendering cost by orders of magnitude compared to volumetric NeRF, and admits a super-resolution upsampler that enables 512×512 training at practical throughput. The triplane design directly inspired the LRM family (§6.2) and remains a dominant architectural building block. StyleNeRF [65] integrates the NeRF volume renderer into a StyleGAN [155] backbone, inheriting its powerful style-based latent structure for fine-grained appearance control, demonstrating that the StyleGAN latent space and NeRF geometry are complementary rather than competing design choices.

A critical limitation of GAN-based approaches is *mode coverage*: the discriminator objective encourages sharp, realistic individual samples but does not explicitly minimise distributional divergence over all plausible views, making it prone to mode dropping for complex, asymmetric scenes. Training instability is a further persistent challenge. These limitations motivated the transition toward diffusion-based approaches, which we discuss next.

7.2 Diffusion Models for Camera-Controlled NVS

Diffusion models [66, 133] have become the dominant generative paradigm for NVS due to three decisive advantages: (i) training stability via a simple denoising objective; (ii) access to powerful image priors from large-scale pretraining; (iii) natural integration of arbitrary conditioning signals (pose, depth, text, reference image) through classifier-free guidance. In contrast to GAN training, which requires careful architectural and hyperparameter tuning to avoid mode collapse, diffusion models can be fine-tuned from large pretrained checkpoints—a substantially lower barrier that has enabled rapid proliferation of camera-conditioned NVS methods.

7.2.1 Pose-Conditioned Single-Image NVS

The most direct application is *single-image NVS*: given one reference photograph and a target camera pose $\boldsymbol{\pi}^*$, generate a plausible rendering at that viewpoint. This is under-constrained without a prior—an infinite number of 3-D scenes are consistent with a single 2-D observation—and diffusion models provide exactly the distributional prior needed to resolve this ambiguity in a perceptually coherent manner.

Zero-1-to-3 and successors. Zero-1-to-3 [130] established the canonical formulation: fine-tune a latent diffusion model (Stable Diffusion [133]) on multi-view renders from Objaverse [37], conditioning on both the reference image and the relative pose $(\Delta\phi, \Delta\theta, \Delta r)$ encoded as a CLIP-space vector. The model learns to "imagine" how an object would look from any viewpoint, without constructing an explicit 3-D representation. The elegance of this formulation lies in its simplicity: by framing NVS as a conditional generation task, it side-steps the need for explicit 3-D supervision while leveraging the strong visual prior of the large diffusion backbone. Zero123++ [147] strengthens multi-view consistency by predicting a fixed grid of six canonical views simultaneously, reducing inter-view inconsistency from independent sampling. One-2-3-45 [109] pipelines Zero-1-to-3 outputs into a cost-volume-based MVS reconstruction [206], converting the diffusion output to a textured mesh within 45 seconds.

Synchronised multi-view denoising. A key challenge with independently conditioned diffusion is that each generated view is sampled from a different realisation of the latent distribution, producing inter-view geometric contradictions. SyncDreamer [204] addresses this by coupling the denoising of multiple target views through shared 3-D volume attention at each diffusion step, enforcing consistent geometry throughout the synthesis process. The 3-D attention mechanism is the architectural mechanism that links the otherwise independent denoising chains, and its introduction marks an important step toward natively 3-D-aware diffusion. MVDiffusion [142] constructs a correspondence-aware attention mechanism that explicitly queries matching pixels across views, enabling holistic panorama generation from a set of overlapping perspective images. Wonder3D [112] takes a two-domain approach: a cross-domain diffusion model simultaneously denoises an RGB and a surface-normal image, with the normal output serving as high-quality geometric supervision for downstream mesh reconstruction. The joint RGB–normal prediction is a principled way to leverage the diffusion model for geometry estimation, exploiting the tight coupling between appearance and geometry that a strong generative prior can encode.

Scalable and pose-free generation. EscherNet [85] generalises pose conditioning via a novel *camera positional encoding* (CaPE), enabling generation of an arbitrary number of views at any resolution without per-angle fine-tuning. This is a significant practical advance: earlier methods required separate models or fine-tuning for each target view configuration. ViVid-1-to-3 [86] recasts NVS as a video generation problem: the temporal axis of a pre-trained video diffusion model is repurposed to model the viewpoint axis, using the model's temporal consistency as a proxy for multi-view coherence. This repurposing leverages the massive scale of video diffusion pretraining for the NVS task, at no additional annotation cost. GeoNVS [28] addresses geometric fidelity by conditioning a video diffusion model on a rendered point cloud at the target pose, grounding the generative hallucination in explicit 3-D geometry.

7.2.2 Multi-View Diffusion and 3-D-Consistent Generation

Single-view methods generate views independently or with limited cross-view coupling. A more ambitious goal is to train a diffusion model that *natively* produces multi-view consistent outputs—effectively encoding 3-D structure implicitly in the model weights. This requires the model to simultaneously denoise all target views while ensuring that the noise-free outputs are consistent renderings of a single 3-D scene.

MVDream [148] achieves this by replacing the temporal self-attention in a video diffusion transformer with a 3-D-aware *multi-view attention* block that attends across all views simultaneously. Training on rendered 3-D assets (Objaverse) with known camera poses teaches the model to share structural information across views, with the multi-view attention serving as the information-sharing mechanism. The resulting model can be used both as a standalone multi-view generator and as a geometry-consistent score function for SDS-based text-to-3D, demonstrating its versatility across both generation paradigms.

GeNVS [23] takes a hybrid approach: a NeRF encoder constructs a 3-D feature volume from the conditioning images, and the diffusion decoder renders a target view by volumetrically querying this feature space. This architecture provides hard geometric grounding while preserving the expressiveness of the diffusion prior for high-frequency appearance details. It represents a principled fusion of reconstruction-based geometry encoding and generation-based appearance synthesis.

7.2.3 Camera Trajectory Conditioning and Video Diffusion NVS

A particularly important frontier is conditioning diffusion models on *continuous camera trajectories*, enabling the generation of smooth, temporally coherent novel-view sequences—the direct bridge to world modeling. A camera trajectory is, in this view, a special case of an action sequence: it specifies how the observer moves through the world over time, and synthesising the corresponding visual experience is the minimal test of a world model's spatial understanding.

Camera trajectory as a control signal. ViewCrafter [215] conditions a video diffusion model (CogVideoX) on point-cloud renderings along a target camera path, using the rendered point cloud as a structural scaffold that prevents temporal drift while the diffusion model fills in appearance and fine geometry. The point-cloud scaffold is a simple but effective way to provide geometric grounding to an otherwise unconstrained video generation model. Free3D [241] injects camera pose into each frame's cross-attention query, enabling a single video diffusion model to follow arbitrary camera trajectories from a single reference image. Novel View Extrapolation [146] specifically targets *large-angle extrapolation*—the regime where geometric hallucination is most necessary—using video diffusion priors to maintain perceptual plausibility when the reference content provides little direct evidence.

Training-free and zero-shot camera control. NVS-Solver [212] demonstrates that an off-the-shelf video diffusion model already encodes sufficient geometric knowledge to solve NVS in a zero-shot manner: by formulating view synthesis as a constrained diffusion process with known-region guidance, it achieves competitive performance without any pose-specific fine-tuning. DDIM Inversion for NVS [197] exploits the deterministic reconstruction property of DDIM [72]: inverting the reference image to its latent code and re-decoding with a modified pose conditioning produces structurally faithful novel views, again without additional training. These training-free approaches reveal an important finding: the geometric knowledge needed for NVS is already latent in large-scale video diffusion models, and the primary task is to elicit it with appropriate conditioning mechanisms rather than to learn it from scratch.

Geometric conditioning for high fidelity. Projected Representation Conditioning [56] proposes projecting 3-D features from the reference view onto the target-view image plane, providing an explicit geometric "warp" prior to the diffusion decoder. This projected feature conditioning substantially reduces hallucination artifacts in regions visible from the target but occluded in the reference. A Single Image and Multimodality [161] augments pose conditioning with depth, surface normal, and semantic maps, each derived from monocular estimation, to enrich the conditioning signal across multiple complementary geometric cues.

7.2.4 Reconstruction–Diffusion Hybrids

A complementary design philosophy *separates* the diffusion and reconstruction stages: diffusion fills in missing views and hallucinations, while a deterministic reconstruction pipeline (NeRF or 3DGS) enforces 3-D consistency. This modular design is attractive because it allows each component to be trained and improved independently, and it provides a clean separation of concerns between generative (appearance hallucination) and geometric (multi-view consistency) reasoning.

Cat3D [47] exemplifies this pattern: given sparse reference views ($n \geq 1$), a multi-view diffusion model generates a dense set of synthetic pseudo-views, which are then fed to a standard 3DGS reconstruction pipeline. Because the diffusion model is fine-tuned jointly with the reconstruction loss, the generated views are explicitly optimised for downstream reconstruction fidelity, not just perceptual quality. ReconFusion [183] integrates the diffusion prior more tightly: it conditions a pixel-diffusion model on pixelNeRF [7] feature queries, acting as a learned regulariser that fills in uncertain regions during sparse-view NeRF optimisation. ZeroNVS [144] handles the challenging case of in-the-wild single images (non-object-centric, background present) by training on scene-level data and conditioning on the camera intrinsics alongside pose, enabling 360° generation from a single casual photograph. NVComposer [101] addresses the case where multiple reference views are available but unposed: a feature composition network aggregates information from all reference images before passing to the diffusion model, improving quality without requiring explicit pose estimation.

7.3 Score Distillation: Diffusion as a 3-D Optimisation Oracle

Score Distillation Sampling (SDS) [13] opened a separate pathway: instead of directly generating novel views, use a pre-trained 2-D diffusion model as a differentiable *score function* to optimise a 3-D representation. The key observation is that the score function of a diffusion model approximates the gradient of the log density of the training data distribution, which provides a signal that pushes rendered images toward the distribution of natural images matching a given text description. The core identity is:

$$\nabla_\theta \mathcal{L}_{\text{SDS}} = \mathbb{E}_{t,\boldsymbol{\epsilon}}\left[w(t)(\hat{\boldsymbol{\epsilon}}_\phi(\mathbf{x}_t; y, t) - \boldsymbol{\epsilon}) \, \frac{\partial \mathbf{x}}{\partial \theta} \right], \tag{5}$$

where θ parameterises the 3-D representation, $\mathbf{x}$ is a rendered image, y is a text prompt, and $\hat{\boldsymbol{\epsilon}}_\phi$ is the diffusion model's noise prediction.

DreamFusion [13] first demonstrated that SDS can synthesise coherent 3-D objects from text descriptions, establishing the proof-of-concept that a 2-D diffusion prior is sufficient to supervise 3-D geometry without any 3-D data. The subsequent SDS literature has largely focused on addressing the two main failure modes of vanilla SDS: *over-saturation* (the generated objects appear too bright and vivid), and *multi-face Janus artifacts* (the model generates separate correct-looking faces on each side of the object rather than a consistent 3-D structure).

Magic3D [17] introduces a coarse-to-fine strategy that first optimises a coarse NeRF then refines a differentiable mesh, achieving $8\times$ higher resolution than DreamFusion by splitting the optimisation into two stages with different representations. ProlificDreamer [175] provides the sharpest theoretical analysis of the over-saturation artifact, identifying it as a consequence of mode-seeking behaviour in the SDS gradient. It proposes *Variational Score Distillation* (VSD), which models the 3-D parameter distribution as a particle set and applies Stein variational gradient descent to obtain diverse, high-quality generation that avoids the mode-seeking bias. Fantasia3D [27] disentangles geometry (optimised via SDS on a DMTet mesh) from appearance (a physically-based BRDF), enabling photorealistic materials and independent editing of shape and texture. GaussianDreamer [211] replaces NeRF with 3DGS as the SDS target, leveraging Gaussian densification for fine detail and achieving real-time rendering of the final asset. LucidDreamer [34] addresses SDS instability with an interval-based score matching scheme that reduces gradient variance.

The SDS paradigm is notable for its complete decoupling of the 3-D representation from the generation model: any differentiable 3-D representation can in principle serve as the target, and any sufficiently capable 2-D diffusion model can provide the score function. This modularity has made SDS a productive research direction, with both the representation and the score function components improving in parallel. The remaining challenge is quality: SDS-based text-to-3D typically produces objects with lower texture quality and fewer fine details than the best feed-forward reconstruction methods, because the SDS gradient is an expectation over all plausible views rather than a direct reconstruction loss.

7.4 Dedicated Multi-View Diffusion Architectures

The methods above predominantly adapt existing video or image diffusion models to the NVS task. A complementary line of work designs architectures *from scratch* for multi-view consistency, taking the view that the temporal attention mechanism of video diffusion models is a suboptimal proxy for the geometric consistency requirements of multi-view generation.

MVDream [148] replaces temporal attention with multi-view attention and trains on Objaverse with ground-truth camera poses, producing natively 3-D-consistent outputs. ImageDream [171] extends MVDream to image-conditioned generation, enabling single-image-to-3D with strong semantic fidelity. Zero123-XL [110] scales the Zero-1-to-3 paradigm to larger model capacities and more diverse training data, demonstrating consistent quality improvements with scale. Direct3D [180] trains a native 3-D latent diffusion model on a triplane latent space, enabling direct 3-D generation without a two-stage pipeline by operating entirely in 3-D feature space.

High-resolution and dense multi-view generation. ERA3D [98] addresses the resolution bottleneck of earlier multi-view diffusion models through a row-wise attention mechanism that efficiently scales to higher-resolution output without the quadratic cost of full 2-D attention. MVDiffusion++ [141] pushes further by using a dense high-resolution attention design that couples overlapping views through a learned correspondence graph, enabling faithful reconstruction from as few as one or two input images. SplatFlow [52] replaces the denoising diffusion process with a multi-view rectified flow formulation, improving both training efficiency and inference speed while maintaining strong 3-D consistency for direct Gaussian splatting synthesis.

Large-scale 3D asset generation. Hunyuan3D [189] and its successor Hunyuan3D 2.0 [229] combine a multi-view diffusion model with a high-fidelity texture synthesis pipeline, scaling text-to-3D and image-to-3D generation to production-quality textured meshes. Their two-stage design—multi-view generation followed by texture baking—provides a clean pipeline architecture that is being widely adopted by downstream production systems. Bootstrap3D [220] addresses the data scarcity problem by synthesising a large-scale multi-view image dataset with a self-training loop, substantially improving the diversity and quality of multi-view diffusion models trained on limited real data. 4Diffusion [53] extends the multi-view paradigm to the temporal dimension, jointly modelling multiple views and multiple timesteps in a single unified 4-D diffusion model.

Video diffusion as a 4-D generation backbone. Stable Video Diffusion (SVD) [15] provides the pre-trained temporal prior that downstream 4-D methods build upon. SV3D [165] fine-tunes SVD on orbital camera trajectories to generate multi-view consistent 3-D objects from a single image. SV4D [194] extends this to dynamic objects by jointly conditioning on camera trajectory and time, synthesising a consistent 4-D video from a single reference. AnimateDiff [51] provides a plug-and-play temporal module for personalised text-to-image models, enabling video generation without per-model video training—a pattern widely adopted in subsequent camera-controlled video diffusion work. CogVideoX [210] scales video generation to high resolution with an expert transformer architecture; its temporal attention blocks have been adopted as a backbone by several NVS methods including ViewCrafter [215] and DimensionX [153]. The Kling model family [84] and Wan [167] represent the latest generation of commercial/open video foundation models that provide strong motion priors for camera-controlled scene synthesis.

Table 4 provides a structured comparison of generative NVS methods across the four paradigms above.

Implications for World Modeling. Diffusion-based NVS represents a critical advance along World Modeling Axis 2 (sparse/single-image generalisation, **D2**) and provides the generative prior essential for completing unobserved scene regions (Axis 1, geometric consistency **D1**). Camera trajectory conditioning (§7.2) directly bridges into Axis 3 (temporal dynamics), as a camera trajectory can be viewed as a special case of a time-ordered action sequence. The remaining gap is *physical action conditioning* (**D4, D5**): existing diffusion NVS methods condition on camera parameters but not on object-level physical interactions, which remains an open problem addressed in §11.

8 Towards Generative World Modeling

Scope and capability tiers. The following discussion focuses on the *visual-generative* branch of world modeling. Methods primarily targeting latent dynamics prediction, model-based reinforcement learning, or planning-centric simulation fall outside our primary scope. Within this scope, we distinguish three capability tiers:

- **Direct world-model instantiations**: methods performing explicit rollout, action-conditioned prediction, or closed-loop simulation (*e.g.*, Genie 2, Cosmos, GAIA-1, UniSim).
- **Bridging methods**: methods connecting 3-D generation with temporal or action-conditioned generation (*e.g.*, ViewCrafter, CAT4D, StreetCrafter).
- **Enabling components**: methods providing geometry, reconstruction, multi-view consistency, or generative priors that world models depend on (*e.g.*, DUSt3R, LRM, Zero-1-to-3, DreamFusion).

The preceding sections have charted the evolution of NVS from per-scene NeRF reconstruction to generative, feed-forward synthesis from a single image. We now address the central thesis of this survey: how does this progression connect to, and ultimately enable, *Generative World Modeling*?

A world model, in the canonical sense introduced by Ha & Schmidhuber [54] and formalised by LeCun [91], is a system that predicts the future state of the world given a current state and an action. Applied to visual, spatial environments, this translates to predicting the *visual experience* of an agent as it moves through and interacts with a scene. One organizing perspective of this survey is that NVS *can be viewed as capturing a restricted special case of world modeling*: it predicts future visual states under camera motion, enforcing geometric consistency without explicit physics. Crucially, this framing has important limits—NVS does not by itself solve object-centric dynamics, action consequences, or physical causality, which remain open challenges addressed in Section 11. The challenge is to extend this capability along three additional axes: temporal dynamics that extend beyond the current frame, agent actions that are richer than camera transforms, and physical plausibility that respects real-world causality.

This section surveys how the field is meeting this challenge across four interconnected directions: 4-D generation and dynamic scene modelling, neural scene simulators for autonomous driving, large-scale world foundation models, and embodied AI applications. We conclude with a formal capability analysis (Table 6) that maps the axis-by-axis progress and identifies the remaining gaps.

8.1 Dynamic Scenes and 4-D World Representation

The step from static NVS to world modeling hinges on modelling *temporal dynamics*: how does a scene evolve over time, and can the model generate novel views at arbitrary future timestamps? The dynamic NeRF and 4DGS methods surveyed in §4–5 address this for *observed* scenes given dense multi-view video. The harder and more practically relevant regime is *generalisable* 4-D generation from sparse or single-image input—precisely the regime inhabited by world models. The key question is whether the powerful 2-D and video priors of large diffusion models can be elevated to serve as 4-D world priors.

Video diffusion models have emerged as the most promising backbone for this purpose, because the temporal axis of a video model naturally encodes the sequential structure of a scene evolving over time. The principal design challenge is coupling the video generation process to spatial geometry, so that the generated frames are not merely temporally coherent but also *multi-view consistent*—a stronger requirement that the pure video generation training objective does not enforce.

ViewCrafter [215] addresses this coupling through a geometric scaffold: it augments a video diffusion model with point-cloud renderings along the target camera trajectory, grounding each generated frame in the known 3-D structure of the reference scene. This scaffold design is principled because it decouples the "what is the geometry" (answered by the point cloud) from the "how does it look" (answered by the diffusion prior), allowing each component to contribute its respective strength. CAT4D [182] generalises the Cat3D framework to the temporal dimension, jointly modelling viewpoint and time with a multi-view video diffusion model, then lifting the output to an explicit 4-D representation; by training jointly over the view and time axes, it learns a shared prior that enforces consistency in both dimensions simultaneously. 4DiM [178] takes a more direct approach, training a 4-D diffusion model that simultaneously denoises a set of frames at varying timestamps and viewpoints, learning a joint spatiotemporal prior without relying on a two-stage pipeline.

SV3D [165] and SV4D [194] extend Stable Video Diffusion to 3-D and 4-D object generation respectively by conditioning on orbital camera paths; the key observation is that the temporal motion prior of a video diffusion model is a strong proxy for the viewpoint change prior of a multi-view model when the conditioning signal specifies the camera trajectory. Consistent4D [76] enforces temporal consistency across a 4-D cascade, resolving inter-frame Gaussian drift that arises when each frame of the 4-D Gaussian sequence is optimised independently. DreamGaussian4D [136] demonstrates that SDS can be applied to 4DGS, enabling text-driven 4-D generation without any video training data, extending the 3-D SDS paradigm to the temporal domain.

Beyond video-diffusion-based approaches, GaussianFlow [126] models 4-D content creation by splatting Gaussian dynamics, providing a principled connection between the explicit Gaussian representation and fluid-like continuous flow dynamics. L4GM [135] scales the LRM paradigm to 4-D, training a large reconstruction model directly on 4-D data to generate consistent dynamic objects from single images in seconds. STAG4D [233] introduces spatial-temporal anchored Gaussian generation that addresses the long-range consistency failures common in 4-D generation by anchoring Gaussian clusters to stable spatial positions. Diffusion4D [104] achieves fast spatial-temporal consistent 4-D generation via a two-stage video diffusion pipeline, reducing the inference time of 4-D generation by an order of magnitude.

A complementary line of work addresses the *monocular* case, where only a single camera view is available throughout time. Diffusion Priors for Dynamic NVS [95] treats the diffusion model as a frame-level regulariser during per-scene 4-D optimisation from a single monocular video, substantially improving novel-view quality in under-constrained regions by providing a learned hallucination prior that fills in views never observed. Dynamic View Synthesis from Small Camera Motion [99] addresses the particularly challenging case where camera motion is small and depth cues are minimal, using a generative prior to resolve the depth-motion ambiguity.

Taken as a whole, the 4-D generation literature demonstrates that the combined signal from spatial consistency (from NVS training) and temporal consistency (from video diffusion training) is sufficient to generate plausible 4-D world states from sparse input. The remaining frontier is *physical grounding*: ensuring that the generated dynamics are not just perceptually coherent but also obey physical laws—a challenge we return to in §11.

8.2 Neural Scene Simulators for Autonomous Driving

Autonomous driving provides a uniquely demanding test-bed for NVS-as-world-model: scenes are large-scale, unbounded, and contain independently moving actors (vehicles, pedestrians) that must be rendered consistently from closed-loop ego-motion trajectories that may deviate substantially from the recorded path. The downstream value of accurate simulation is concrete and measurable: closed-loop evaluation of autonomous systems requires the ability to replay any scenario with arbitrary modifications to weather, traffic density, and actor behaviour. Three architectural strategies have emerged, representing progressively stronger integration between scene representation and generative modelling.

NeRF-based compositional simulators. The first strategy separates the scene into independently-modelled components that can be edited and composited. MARS [186] separates the scene into a static NeRF background and per-agent NeRF foregrounds, allowing independent actor manipulation—an approach that directly addresses the compositionality requirement of scene editing but requires per-instance annotations. UniSim [223] extends this to a closed-loop sensor simulator that supports controllable scene editing (removing actors, changing weather, inserting novel agents) with photorealistic rendering at the ego sensor, demonstrating that a compositional NeRF scene model can serve as a full sensor simulation system. EmerNeRF [209] removes the requirement for per-actor annotations by learning the dynamic-static decomposition in a self-supervised manner from scene flow estimated via multi-camera video; this is an important step toward scalable scene decomposition that does not require expensive human labelling.

3DGS-based driving reconstruction. The second strategy replaces NeRF with Gaussian Splatting, inheriting its real-time rendering advantage, which is critical for closed-loop simulation at scale. Street Gaussians [208] extends 3DGS to urban scenes by representing tracked foreground objects with separate Gaussian models and compositing them over a background Gaussian field; the per-object Gaussian decomposition supports instance-level editing without re-training the full scene model. PVG [30] models Gaussian motion with periodic vibration functions that efficiently encode the slow, cyclic motions common in urban traffic, such as the repetitive movement of wipers or the oscillation of pedestrian gaits. OmniRe [33] unifies background, foreground, and sky in a single Gaussian framework, establishing a strong all-in-one baseline that subsumes the separate components of earlier compositional approaches. DrivingGaussian [244] addresses large-scale urban environments by compositing incremental static and dynamic Gaussian fields with a composite dynamic Gaussian graph for multi-camera consistency. NeuRAD [163] and SplatAD [60] further scale Gaussian-based driving simulation to production-grade multi-sensor (LiDAR plus camera) settings, demonstrating that the representation can accommodate the full sensor suite of a commercial autonomous vehicle.

Generative world models for autonomous driving. The third strategy transcends deterministic scene reconstruction by training *generative* models that can produce novel, physically plausible driving scenarios beyond the recorded dataset. This capability is essential for safety validation, which requires testing on corner cases that are rare or absent in recorded data.

DriveDreamer [187] is the pioneering work in this direction, explicitly framing autonomous driving scene generation as a world model problem by training a diffusion model conditioned on structured traffic annotations and ego-motion to produce photorealistic, temporally coherent multi-camera video. DriveDreamer-2 [45] scales this approach with LLM-generated traffic layouts, enabling greater diversity and user-controllable scenario editing without re-training, by treating the LLM as a compositional scene language that the diffusion model can interpret. Drive-WM [205] couples scene prediction with planning, demonstrating that a world model trained on real driving data can directly improve downstream planning metrics—a result that validates the practical value of world model training for the end task of safe navigation. OccWorld [166] takes a 3-D occupancy perspective, learning a world model that predicts future occupancy grids jointly with ego-motion, unifying scene forecasting and motion planning in a single autoregressive framework that operates in explicit 3-D space rather than image space. WoVoGen [70] addresses the multi-camera consistency challenge by reasoning over a shared 3-D world volume, ensuring that all generated camera views are geometrically coherent—a non-trivial constraint when cameras have limited overlap.

Complementing these end-to-end world models, several methods address specific components of the simulation pipeline. StreetCrafter [207] uses a point-cloud-conditioned video diffusion model with controllable camera motion for data augmentation in street-view synthesis, bridging the generation and reconstruction paradigms. Sat2Vid [184] lifts a satellite image to a street-view panoramic video, addressing the altitude gap between aerial mapping data and ground-level perception—a capability that could enable training data generation for any mapped region without physical vehicle traversal. Wild-GS [198] solves the unconstrained internet-photo case by disentangling transient and persistent Gaussian appearance in 3DGS, enabling reconstruction from tourist photos with varying weather and crowds.

The driving simulation literature illustrates the full range of the NVS-to-WM transition: at one end, reconstructive methods (MARS, OmniRe) provide accurate but bounded-to-observed-data replays; at the other, generative world models (DriveDreamer, OccWorld) provide controllable synthesis of novel scenarios. The ideal system would combine the geometric accuracy of reconstruction with the diversity and controllability of generative models—an active open problem.

8.3 Generative World Models and Foundation Models

World foundation models represent the apex of the NVS-to-world-modeling trajectory: systems that jointly handle viewpoint, time, and action, trained at internet scale. These systems are distinguished from domain-specific simulators by their breadth: they are designed to generalise across diverse environments and task domains rather than to achieve maximum fidelity in a narrow setting. We identify three design axes that differentiate current approaches.

Sequence vs. spatial representation. The first design axis concerns whether the world state is represented as a flat sequence of tokens or as a spatially structured representation. Flat-sequence models are simpler to train at scale but sacrifice explicit geometric structure; spatial models are harder to train but support view-consistent generation by construction.

GAIA-1 [61] treats driving scenes as sequences of video, text, and action tokens, training an autoregressive transformer to predict the next token. This approach achieves strong temporal coherence but lacks an explicit 3-D representation, limiting viewpoint flexibility to the trained camera configurations. Genie [16] learns a latent action space from unsupervised internet videos, enabling interactive game-world generation without action labels—a significant step toward self-supervised world model pretraining that does not require expensive action annotation. The latent action discovery capability of Genie is particularly noteworthy because it demonstrates that world model training does not require labelled action data, which is scarce, but can exploit the vastly more abundant unlabelled video data. WorldDreamer [188] adopts a masked-token prediction objective over a joint video-text tokenisation, demonstrating that masked world model pre-training transfers well to diverse downstream generation tasks—analogously to how BERT-style pre-training transfers to NLP tasks.

3-D-grounded generation. Genie 2 [123] introduces persistent 3-D structure into the Genie framework, generating scenes that maintain geometry across viewpoints and support first-person interactive navigation—currently the closest system to an NVS-grounded world model for open-ended environments. Its ability to generate geometrically consistent first-person view sequences in novel environments from a single image is a striking demonstration of the power of combining video diffusion with spatial grounding. WonderWorld [214] takes a complementary approach: instead of learning from large-scale video, it uses instant 3DGS initialisation from a single image combined with diffusion inpainting to generate interactive 3-D scenes on the fly, demonstrating that a lightweight system can produce interactive world experiences without any world model pretraining. WonderJourney [213] chains text-guided scene generation steps to produce perpetually explorable 3-D environments, addressing the long-horizon generation challenge by breaking it into sequential local steps. Wonderland [105] enables navigation through 3-D scenes from a single image by combining monocular depth estimation with video diffusion, demonstrating that the combination of geometric lifting and generative inpainting can produce compelling spatial experiences. AVID [116] adapts pre-trained video diffusion models to serve as general-purpose world models for offline reinforcement learning, demonstrating the deep connection between video generation and world modeling: a model trained for visual prediction can directly serve as an environment model for policy learning without task-specific world model training. Structured World Models [132] learn structured representations from human videos in a fully unsupervised manner, enabling policy learning from video without any action labels.

Physical-world foundation models. Cosmos [9] is the most ambitious current effort: a family of world foundation models pretrained on internet-scale video with explicit physical world conditioning, designed for downstream fine-tuning to robotics and autonomous driving. Its training methodology—combining large-scale video pretraining with physics-informed conditioning—represents the field's most direct attempt to learn physical world dynamics from observation alone. Pandora [191] interleaves language and video tokens, enabling language-driven world simulation with both visual and linguistic output; the joint language–video generation capability is essential for instruction-following world models that must respond to natural language commands. The *Is Sora a World Simulator?* analysis [230] provides a comprehensive assessment of the relationship between video generation models and world simulation capability, identifying key gaps in physical consistency, causal understanding, and controllability that remain open in current video foundation models. This gap analysis is directly relevant to our survey because it quantifies how far the current best video generation systems fall short of a full world model.

The NVS-to-WM transition. A key architectural trend is the convergence of NVS and WM techniques. Whereas early world models like GAIA-1 operated in 2-D video space, recent systems increasingly incorporate explicit 3-D representations—whether Gaussian splats, triplane NeRFs, or occupancy grids—as the world state that the generative model predicts and renders. This convergence has both practical and theoretical significance. First, 3-D-aware world models naturally support multi-camera and novel-viewpoint prediction, which is essential for robotic and driving applications where the sensor configuration may change at test time. Second, the NVS community's work on generalisable reconstruction provides the initialisation and geometry backbone that world models need to maintain spatial consistency. Third, the world model community's work on temporal prediction and action conditioning provides the dynamics prior that pure NVS systems lack. Table 6 captures this convergence precisely via the dual axis of *spatial generalisation* (D1–D2) and *temporal/action conditioning* (D3–D5).

Table 5 compares world modeling methods.

8.4 Embodied AI and Robotics

NVS is more than a rendering tool in robotics: it provides the *spatial prior* that allows agents to reason about unseen parts of the environment before taking action. This is precisely the world-model capability that matters for robotic manipulation and navigation—the ability to predict "what would I see if I moved there?" without physically executing the motion. We identify three use patterns that span the spectrum from pure pose estimation to full sim-to-real transfer.

Pose estimation and object localisation. iNeRF [87] inverts a trained NeRF to estimate 6-DoF camera pose by gradient descent on the rendered photometric error—the first demonstration that NeRF's differentiable renderer can be repurposed for *inference* (estimating the state of the world) as well as reconstruction (building a model of it). This inversion paradigm anticipates the broader use of differentiable neural renderers for perception tasks. Dex-NeRF [67] applies NeRF to transparent-object grasping, where conventional depth sensors fail; by leveraging NeRF's ability to model view-dependent appearance, it infers geometry from specular reflections that are invisible to depth cameras.

Visual planning and manipulation. NeRF-Supervisor [232] uses a pre-trained NeRF to synthesise robot-wrist camera views at hypothetical future poses, enabling visual planning without physical rollouts—a direct application of NVS as a world model for robot manipulation. GaussianGrasping [149] replaces NeRF with 3DGS for real-time manipulation planning, exploiting the explicit Gaussian representation for fast collision checking and grasp pose optimisation; the real-time rendering speed of 3DGS is essential for reactive planning in dynamic environments.

Simulation-to-real transfer. RoboGSim [97] reconstructs robot workspaces in 3DGS and renders synthetic training demonstrations from novel viewpoints, directly closing the sim-to-real gap for visuomotor policy learning. By generating photorealistic training data at arbitrary viewpoints and configurations—including viewpoints that were never physically observed—it demonstrates that a 3DGS world model can serve as a data augmentation engine for downstream policy learning.

The robotic applications of NVS reveal the concrete value proposition of geometric world models: they enable agents to reason about unseen viewpoints and states without physical exploration, reducing the cost and risk of real-world data collection. This is the most direct current application of world modeling to embodied AI, and it points toward a future in which robots continuously update and query their spatial world models as part of closed-loop task execution.

8.5 Human Novel View Synthesis

Synthesising novel views of humans is a critical subproblem with applications in AR/VR, telepresence, and embodied AI. The challenge is distinct from general scene NVS because human bodies have a known structural prior (the kinematic skeleton and body shape parameterised by SMPL) that can serve as a geometric scaffold, but also exhibit complex non-rigid deformation (clothing, hair, facial expression) that the scaffold alone cannot represent.

Neural implicit human models. Neural Body [125] was the first work to combine the SMPL structural prior with NeRF, anchoring structured latent codes on SMPL vertices to enable novel view and pose synthesis with strong body geometry constraints. Rotationally-Temporally Consistent [111] enforces both rotation and temporal consistency for human performance video synthesis, addressing the specific failure mode of per-frame NeRF methods that produce flickering or geometrically inconsistent results across frames. NHR [181] adopts multi-view point-cloud-based rendering for dynamic human synthesis, demonstrating that explicit point representations can support high-quality human rendering before the 3DGS era.

Generalisable and diffusion-based human NVS. CFSynthesis [102] enables controllable and free-view 3-D human video synthesis via a disentangled representation that separates body pose, shape, and appearance, enabling independent control of each. iVS-Net [31] learns human view synthesis from internet videos without any 3-D supervision, demonstrating that the abundance of 2-D human video on the internet is sufficient to learn a strong human NVS prior without expensive multi-view capture studios. Diffuman4D [242] achieves 4-D consistent human view synthesis from sparse-view videos via video diffusion, extending the video-diffusion-as-4D-prior paradigm to the human domain. EVA-Gaussian [62] achieves real-time 3DGS-based human NVS under diverse multi-view settings, combining the real-time rendering speed of 3DGS with a feed-forward generalisation architecture for fast inference on unseen subjects. Blur2Sharp [39] addresses human novel pose and view synthesis from blurry input with a generative prior, targeting the practical setting where captured reference images have motion blur.

Human NVS methods represent an important special case of the general world modeling problem: the human body is a system with known structure, highly constrained dynamics (governed by skeletal kinematics), and rich appearance variation (clothing, lighting, expression). The success of structured human models suggests a general design principle for world modeling: incorporating domain-specific structural priors substantially reduces the burden on the learned generative component.

8.6 The Boundary: NVS vs. World Modeling

Table 6 delineates the capability boundary between NVS and full World Modeling, using the five desiderata (D1–D5) introduced in §2. The table reveals that contemporary NVS systems have substantially closed gaps D1 and D2—geometric consistency and sparse-input generalisation—but that D3 (temporal dynamics), D4 (action conditioning), and D5 (physical plausibility) remain only partially addressed even by the most advanced world foundation models. This analysis defines the open research frontier we discuss in §11.

9 Datasets and Benchmarks

Empirical progress in NVS and world modeling is inseparable from the availability of suitable datasets. A training or evaluation dataset defines the implicit scope of the problem: it determines which scene types, capture modalities, viewpoint distributions, and annotation types a method must handle. The field has developed a rich ecosystem of datasets spanning a wide range of domains—from controlled synthetic objects to urban driving sequences—and understanding their respective strengths and limitations is essential for interpreting benchmark results. We organise the dataset survey along four axes: object-centric datasets for single-image and multi-view reconstruction, static scene datasets for per-scene NeRF and 3DGS evaluation, dynamic scene datasets for temporal NVS, and large-scale autonomous driving datasets for neural scene simulation. Table 7 at the end of this section provides a structured comparison.

9.1 Object-Level Datasets

Object-centric datasets are the foundation of the generative 3-D literature, because they provide clean, well-posed reconstruction problems with known geometry and camera parameters, enabling rigorous evaluation of single-image and sparse-view reconstruction methods.

ShapeNet [24] established the canonical benchmark for single-image 3-D reconstruction with its collection of 51K CAD models across 55 object categories. Its clean geometry and known cameras make it ideal for controlled evaluation, though the synthetic rendering distribution departs substantially from real-world photograph appearance. **Objaverse** [37] resolved the scale bottleneck by providing over 800K 3-D objects sourced from the internet, enabling the large-scale pretraining required by LRM-family methods that would overfit on ShapeNet's limited diversity. **Objaverse-XL** [38] further scales to 10M+ assets, providing the data regime required by the next generation of foundation reconstruction models. **Co3D** [71] provides 19K+ annotated per-category real video sequences, enabling evaluation of category-level NVS at a much higher realism than synthetic benchmarks; its structured train/test splits support rigorous evaluation of generalisation within and across categories. **MVImgNet** [216] adds 238K multi-view real-image sequences across 238 categories, bridging the synthetic-to-real gap while maintaining the structured multi-view annotation needed for NVS evaluation.

9.2 Static Scene Datasets

Static scene datasets test the fidelity and generalisation of per-scene and generalisable NVS methods on real-world complexity, without the confounding factor of temporal dynamics. They span the full range of scene types encountered in practice, from controlled tabletop objects to unbounded outdoor environments.

NeRF-Synthetic [11] provides 8 synthetic bounded objects with known geometry and high-quality Blender renders, making it the canonical baseline for evaluating per-scene NeRF methods under controlled conditions. **LLFF** [12] provides 8 real forward-facing scenes captured with a handheld camera, testing methods' ability to handle real image noise, exposure variation, and limited view range. **DTU** [129] offers 124 controlled multi-view stereo scenes with ground-truth point clouds, enabling combined NVS and geometry evaluation. **Tanks and Temples** [5] provides large-scale indoor and outdoor scenes captured with professional cameras, testing scalability to scenes that are substantially larger and more complex than tabletop objects. **Mip-NeRF 360** [73] is the current gold standard for unbounded 360° evaluation, with 9 scenes carefully selected to represent diverse outdoor and indoor environments with non-trivial background depth. **ScanNet** [3] provides 1513 indoor RGB-D sequences with semantic and instance labels, serving as the primary benchmark for indoor NVS methods that must handle the large viewpoint variation and partial occlusion typical of room-scale captures. **ScanNet++** [19] upgrades this to DSLR-quality imagery alongside iPhone captures, enabling evaluation of high-fidelity indoor NVS models. **RealEstate10K** [158] provides 10K real estate walkthrough videos with estimated camera trajectories, serving as the primary training and evaluation set for generalisable video NVS methods that must handle large camera motions and diverse indoor scenes.

9.3 Dynamic Scene Datasets

Dynamic scene datasets are essential for evaluating methods that target the temporal generalisation requirements of world modeling. They expose the additional challenges of large non-rigid deformation, occlusion handling under motion, and the depth-motion ambiguity that plagues monocular dynamic reconstruction.

D-NeRF [6] provides 6 monocular synthetic dynamic objects with known deformation fields, enabling quantitative evaluation of dynamic reconstruction methods under controlled conditions. **HyperNeRF** [79] provides 15 real monocular sequences of objects undergoing topological changes (opening a noodle box, cutting a cucumber), testing methods' ability to represent topology-changing dynamics that smooth deformation fields cannot model. **NHR** [181] and **CMU Panoptic** [77] provide multi-view captures of human performance, with Panoptic offering 65 sequences of social activities from a dense 31-camera array—the richest multi-view human performance dataset publicly available. **Neu3D** [219] provides 6 high-quality multi-view human video sequences that have become the standard benchmark for multi-view dynamic NeRF evaluation. **iPhone Dataset** [50] provides 14 casually-captured phone video sequences of dynamic scenes, testing methods' ability to handle the most challenging regime: monocular capture with arbitrary camera and scene motion.

9.4 Autonomous Driving and Large-Scale

Autonomous driving datasets provide the large-scale, sensor-rich captures needed to evaluate neural scene simulators and driving-domain world models. They are distinguished from the above categories by their multi-sensor modality (cameras plus LiDAR), large-scale sequential capture, semantic and instance annotations, and ego-motion ground truth.

Waymo Open [152] provides 1000 multi-camera plus LiDAR driving segments covering diverse geographic and weather conditions, serving as the primary evaluation benchmark for neural driving simulators. **nuScenes** [20] adds 360° camera coverage and HD maps to 1000 driving scenes, enabling evaluation of methods that must maintain consistency across the full surround-view camera ring. **KITTI-360** [203] provides urban driving sequences with panoptic segmentation labels, enabling combined NVS and semantic evaluation. **Argoverse 2** [179] provides 1000 high-resolution 360° driving sequences with precise sensor calibration and HD map annotations, supporting the evaluation of large-scale neural scene simulators at the current frontier of sensor quality. **PandaSet** [192] provides 103 scenes with a forward-facing LIDAR plus multiple cameras, useful for evaluating fusion of camera and LiDAR NVS in a single consistent benchmark.

Table 7 provides a structured comparison of key datasets.

10 Evaluation Metrics and Protocols

Evaluation in NVS and world modeling is more nuanced than in many areas of computer vision, because the goals of the field span a spectrum from perceptual realism to geometric accuracy to physical plausibility—and no single metric captures all three simultaneously. A systematic understanding of the strengths and limitations of existing metrics is essential for interpreting the benchmark tables in this survey and for identifying where reported numbers may be misleading. We first discuss the standard image-quality metrics that dominate the per-scene NVS literature, then address the video-quality and task-specific metrics required for world modeling evaluation, and finally identify the evaluation gaps that represent open methodological problems.

10.1 Standard Image Quality Metrics

The three metrics that appear in virtually all NVS benchmarks measure different aspects of the relationship between a synthesised novel view $\hat{\mathbf{I}}$ and a ground-truth photograph $\mathbf{I}^*$.

PSNR (Peak Signal-to-Noise Ratio) measures pixel-level fidelity as a logarithmic function of mean squared error: $\text{PSNR} = 10 \log_{10}(1/\text{MSE})$. High PSNR indicates low pixel-level error, but the metric is insensitive to structural artifacts and perceptually meaningless differences. It is nonetheless a standard baseline because it requires no learned components and is fully deterministic.

SSIM [221] (Structural Similarity Index) computes local statistics—mean, variance, and covariance—in sliding windows across the image, capturing structural coherence beyond pixel-level fidelity. It correlates better with human perceptual quality than PSNR in the classical reconstruction regime, but like PSNR it penalises perceptually plausible hallucinations that happen to be pixel-misaligned with the ground truth.

LPIPS [238] (Learned Perceptual Image Patch Similarity) measures perceptual similarity via deep feature distances, computing the ℓ_2 distance between activations of a pretrained AlexNet or VGG at multiple layers. It is the metric most closely correlated with human perceptual judgements and is therefore the most informative single metric for evaluating photorealistic NVS quality. The limitation of LPIPS is its learned nature: performance on out-of-distribution scenes (e.g., medical imagery, infrared) may not reflect perceptual quality.

Table 8 shows representative benchmark results on NeRF-Synthetic and LLFF across the major method families.

10.2 Video and World Modelling Metrics

The transition to world modeling requires metrics that evaluate temporal and dynamic properties in addition to per-frame image quality. Single-frame quality metrics are insufficient for this purpose because they do not measure whether adjacent frames form a geometrically and temporally coherent sequence.

FVD [157] (Fréchet Video Distance) measures the distributional distance between generated and real video sequences in the feature space of a pretrained I3D network, capturing both spatial quality and temporal dynamics. It is the video-generation analogue of FID [115] and has become the primary metric for evaluating world model generation quality. Its limitation is that it measures distributional quality rather than per-instance fidelity—a model that generates diverse but geometrically inconsistent sequences can achieve a good FVD score.

Temporal consistency is typically measured as the average cosine similarity of DINO [] features between consecutive frames, penalising large changes in feature space that correspond to flickering or frame-level inconsistency. This metric is sensitive to the temporal coherence that is most directly relevant to world modeling but does not capture geometric 3-D consistency across viewpoints.

Task-specific evaluation is the most informative metric for world models but also the most expensive to compute. For autonomous driving world models, the key task-specific metrics are closed-loop collision rate, route completion ratio, and imitation gap (the difference between the model and an expert policy in a closed-loop setting). For robotics, task success rate and generalisation to novel configurations are the primary metrics. These task-specific metrics provide the most direct evidence that a world model is learning semantically useful scene dynamics rather than merely reproducing visual appearance statistics.

10.3 Limitations of Current Evaluation

The gap between current evaluation protocols and the requirements of world modeling represents a major methodological problem that the field has not yet fully addressed.

PSNR and SSIM penalise perceptually plausible but pixel-misaligned generations, particularly under large viewpoint changes where the reference content provides little direct evidence for the target-view appearance. This creates a systematic bias against generative methods, which often hallucinate content that is physically plausible but does not match the ground-truth pixel values used for evaluation. FVD captures temporal dynamics but not geometric accuracy, allowing methods that generate temporally smooth but spatially inconsistent videos to achieve high scores. As a result, no existing benchmark jointly evaluates geometry, dynamics, action responsiveness, and physical plausibility—the full set of world-modeling desiderata. This evaluation gap is itself a major open problem.

Towards systematic evaluation. The work on Systematic Evaluation of NVS [25] provides a principled analysis of how existing benchmarks differ and proposes controlled evaluation protocols separating appearance and geometry quality. Non-Aligned Reference IQA [239] addresses the fundamental problem of quality assessment when reference and synthesised views are not pixel-aligned, proposing a learned non-aligned metric for NVS that is better suited to the large-viewpoint-change regime where generative methods operate. VGGT-X [108] extends VGGT to dense NVS, providing strong baselines for the emerging class of feed-forward 3-D foundation models and enabling consistent comparison across the generalised reconstruction and generative NVS families. We propose in §11 a five-axis world-modeling benchmark that would provide the systematic, multi-dimensional evaluation that the field currently lacks.

11 Open Problems and Research Roadmap

The survey presented in the preceding sections makes clear that the field of NVS-to-world-modeling is progressing rapidly along multiple fronts simultaneously. Per-scene reconstruction from dense multi-view input has become comparatively mature under controlled benchmarks, though open challenges remain in scenes with reflective or transparent surfaces, dynamic lighting, very large scale, and out-of-distribution appearance. Feed-forward generalisation to sparse-view and single-image input has seen dramatic advances in the past two years. And the first generation of world foundation models trained at internet scale has demonstrated that video generation and spatial synthesis are convergent problems. Yet the capability table in Section 8 (Table 6) reveals that three of the five world-modeling desiderata—temporal dynamics (D3), action conditioning (D4), and physical plausibility (D5)—are only partially satisfied by current best methods. This section identifies the most important open problems that must be resolved to cross these frontiers, and proposes a concrete research roadmap.

11.1 The Geometry–Generation Trade-Off

The fundamental tension between the *reconstruction* and *generative* paradigms has not yet been resolved. Reconstruction-based methods (NeRF, 3DGS, feed-forward Gaussian) deliver geometric precision but are limited to what has been observed: they cannot hallucinate content outside the observed view frustum. Generative methods (diffusion NVS, world models) provide rich priors over plausible scene completions but can violate multi-view geometric consistency, producing views that are individually realistic but mutually contradictory in 3-D space.

Methods like ReconFusion [183], Cat3D [47], and ViewCrafter [215] begin to bridge this gap by coupling a generative prior to an explicit 3-D scaffold, but the integration is primarily architectural (conditioning the diffusion model on a point cloud or NeRF) rather than principled (ensuring that the generated content satisfies 3-D consistency constraints during denoising). Principled solutions—such as geometry-grounded diffusion that enforces epipolar constraints during denoising, or diffusion-seeded 3DGS that initialises the Gaussian field from a generated multi-view set—represent an active frontier with significant room for innovation. The key insight we advocate is that the 3-D consistency constraint should be enforced *during* generation rather than as a post-hoc filter, which would require integrating differentiable rendering into the diffusion denoising loop.

11.2 Long-Horizon Temporal Coherence

Existing 4-D methods maintain coherence over a few seconds, corresponding to a few hundred frames of video. World models for autonomous driving and robotics require coherence over minutes of operation, during which the scene may change substantially due to agent interactions, illumination changes, and new scene content entering the field of view. Two promising directions have been identified in the recent literature, though neither has yet achieved the long-horizon coherence required by real-world applications.

The first direction is *persistent scene memory*: Gaussian maps that accumulate observations across time and preserve global consistency by storing the entire world state as a growing set of Gaussian primitives. Street Gaussians [208] and OmniRe [33] demonstrate that this approach can handle scene-length (several-minute) driving sequences, but their memory grows without bound and they do not support closed-loop rollouts of novel trajectories. An efficient, bounded-memory Gaussian world state that can be queried and updated in real time remains an open problem.

The second direction is *world model pretraining*: leveraging internet-scale video prediction as a temporal prior that can be fine-tuned to specific domains. Cosmos [9] and Genie 2 [123] represent the current frontier here, but their coherence degrades over longer horizons due to accumulated prediction errors in the autoregressive generation process—a fundamental challenge that may require non-autoregressive or hierarchical temporal architectures.

11.3 Physical Grounding

Current methods learn the appearance correlations that arise from physical laws, but they do not represent physical laws explicitly. This distinction matters for world modeling: a model that has learned that objects tend to remain stationary may generate plausible-looking video, but it will fail to predict the correct response to an unusual physical interaction (pushing an object off a table, flooding a scene, deforming an elastic material). Physical grounding is therefore a prerequisite for reliable world models in applications that require reasoning about novel or unusual physical events.

PAC-NeRF [100] demonstrates a promising direction by integrating continuum mechanics into NeRF, enabling system identification of material properties from observed deformation. PhysGaussian [193] brings physics simulation to Gaussian Splatting, embedding Material Point Method (MPM) simulation directly in the Gaussian optimisation loop. The central challenge is extending these approaches to complex, open-world scenes with diverse materials and contact interactions: both PAC-NeRF and PhysGaussian have been demonstrated primarily on single-material, simple-geometry scenarios. Integrating differentiable physics more generally into the rendering pipeline in a way that scales to complex scenes and diverse material properties remains a largely open problem.

11.4 Open-Vocabulary Semantic Integration

The methods surveyed in this paper largely address appearance-level scene understanding, but a full world model must also reason about *semantics*: what is this object, what can I do with it, how will it respond to my action? The recent language-grounded 3-D field represents the most direct progress toward this goal. LERF [68] grounds CLIP features in 3-D via NeRF, enabling open-vocabulary 3-D queries such as "where is the wooden chair?". LangSplat [128] brings language grounding to 3DGS, achieving real-time semantic query response while maintaining the rendering speed advantage of the Gaussian representation. Feature3DGS [243] provides a more general framework, distilling arbitrary 2-D foundation model features—including SAM [83] and DINO—into the Gaussian representation.

Extending these language-grounding approaches to dynamic, actionable scenes is a key step toward language-grounded world models: a model that can both generate novel views of a scene and answer natural language queries about what it contains, how it will evolve, and how an agent can interact with it. This integration of spatial and semantic reasoning is a central challenge for the next generation of world models.

11.5 Evaluation and Benchmarks

The evaluation gap identified in Section 10 requires a new benchmark design that simultaneously tests all five world-modeling desiderata. We propose a **World Modeling Evaluation Suite** with five axes:

(i) *3-D consistency* measured by multi-view photometric consistency under adversarial viewpoints that maximally stress the geometric model;

(ii) *Temporal coherence* measured over long horizons ($>$30 s) using feature-space consistency and structural similarity;

(iii) *Action responsiveness* measured in interactive settings by the quality of generated responses to specified agent actions;

(iv) *Physical plausibility* verified by comparison against physics-engine simulation of the same scenario;

(v) *Open-world generalisation* measured by single-image synthesis quality across highly diverse, out-of-distribution environments.

No existing benchmark covers all five axes. We note that designing such a benchmark is itself a substantial research contribution, as it requires defining a controlled test-bed for all five dimensions without introducing confounding factors between them.

11.6 Research Roadmap

Drawing the threads of this analysis together, we propose five research directions as the most productive frontiers for the immediate future.

R1. Unified geometry-generation architectures. Develop architectures that couple explicit 3-D representations (NeRF, 3DGS) with diffusion priors in a manner that enforces 3-D consistency during generation, not merely as a post-hoc filter. The key technical challenge is integrating differentiable rendering into the denoising loop without prohibitive computational cost.

R2. Internet-scale feed-forward generalisation. Scale feed-forward reconstruction models (LRM, DUSt3R, VGGT) to internet-scale pretraining on diverse unstructured video, enabling open-world NVS from a single image with generalisation to arbitrary scene types encountered at deployment time.

R3. Differentiable physics in 4-D world models. Integrate differentiable physics priors into 4-D world model training, building on the PAC-NeRF and PhysGaussian foundations, to enable physically grounded prediction of novel physical interactions.

R4. Language and semantic grounding for action conditioning. Extend language-grounded 3-D representations (LERF, LangSplat, Feature3DGS) to dynamic, actionable world models that can respond to natural language action specifications, enabling instruction-following spatial agents.

R5. A comprehensive world modeling evaluation suite. Design and release a benchmark covering all five world-modeling desiderata, with controlled, reproducible evaluation protocols that separate each axis's contribution to total system performance. This benchmark is a prerequisite for measuring progress on the most challenging open problems.

Together, these five directions define a research agenda that, if successfully prosecuted, would complete the transition from current NVS systems—which solve D1 and partially D2—to full world models that satisfy all five desiderata.

12 Conclusion

This survey has traced the evolution of Novel View Synthesis from its origins in image-based rendering through the neural radiance field era, the Gaussian Splatting revolution, the emergence of generalisable feed-forward reconstruction, and the convergence with generative world modeling. The organizing perspective we have pursued—that NVS provides essential geometric building blocks for the visual-generative branch of inference engine of Generative World Modeling—finds strong empirical support in the recent literature, where the most capable world models (Genie 2, Cosmos, ViewCrafter) are distinguished from their predecessors precisely by their explicit incorporation of spatial consistency constraints derived from the NVS research tradition.

The unified continuous-spectrum taxonomy we have proposed organises more than 180 methods along three axes (input regime, temporal scope, controllability) and makes explicit their world-modeling implications. Viewed through this lens, the history of the field is a systematic expansion of coverage along the spectrum: from dense multi-view static reconstruction at the origin, through sparse-view and single-image generalisation in the middle range, toward the 4-D, action-conditioned, physically plausible generation that defines the far end of the spectrum and the open research frontier.

The field is at a genuine inflection point. The spatial grounding developed by the NVS community and the temporal and action capabilities developed by the world modeling community are actively converging, as evidenced by systems such as Genie 2, Cosmos, ViewCrafter, and the neural driving simulators reviewed in Section 8. This convergence is not merely a metaphor: architecturally, the most capable recent systems are genuine fusions of explicit 3-D representations and large-scale video generation priors, combining the geometric accuracy of the former with the coverage and controllability of the latter.

Closing the geometry–generation trade-off, achieving long-horizon temporal coherence, and grounding predictions in physical laws are the three primary technical challenges ahead. The five research directions proposed in Section 11 represent our assessment of the most productive paths to addressing these challenges. Progress along these directions will not only advance the field of NVS and world modeling in isolation, but will also enable downstream capabilities in autonomous driving, robotic manipulation, and interactive AI agents that require the ability to reason about and generate plausible futures of the physical world.

The survey is an ongoing work. Content will be further expanded and updated as the field continues to evolve at its current rapid pace.

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Table 4: Generative NVS methods. **Backbone**: LDM=latent diffusion, VDM=video diffusion, GAN=generative adversarial. **Ctrl**: pose conditioning mechanism. **3D Out**: whether the method produces an explicit 3-D representation.

Method	Venue	Backbone	Ctrl	3D Cons.	3D Out	Key Contribution
GAN-based						
GRAF [80]	NeurIPS'20	NeRF-GAN	Latent	∼	NeRF	Radiance field GAN
pi-GAN [41]	CVPR'21	NeRF-GAN	Latent	∼	NeRF	Periodic implicit
GIRAFFE [120]	CVPR'21	Feat. Field	Latent	∼	Neural	Compositional
EG3D [22]	CVPR'22	Triplane	Latent	✓	Tri.	Triplane+SR
StyleNeRF [65]	ICLR'22	StyleGAN	Latent	∼	NeRF	Style-based NeRF
Pose-conditioned single-image diffusion						
Zero-1-to-3 [130]	ICCV'23	LDM	Δpose	∼	–	Pose-cond LDM
Zero123++ [147]	arXiv'23	LDM	Δpose	∼	–	6-view grid
One-2-3-45 [109]	NeurIPS'23	LDM+MVS	Δpose	✓	Mesh	Diff$\rightarrow$MVS
SyncDreamer [204]	ICLR'24	LDM	Δpose	✓	–	Volume sync
Wonder3D [112]	CVPR'24	LDM	Δpose	✓	–	RGB+Normal
EscherNet [85]	CVPR'24	LDM	CaPE	✓	–	Arbitrary #views
ViVid-1-to-3 [86]	arXiv'24	VDM	Δpose	∼	–	Video=NVS axis
Camera trajectory / video diffusion						
Free3D [241]	ICLR'24	VDM	Cam embed	∼	–	Per-frame pose
ViewCrafter [215]	arXiv'24	VDM	Ptcloud	✓	–	Scaffold+VDM
NVS-Solver [212]	arXiv'24	VDM	Zero-shot	∼	–	Guided diffusion
DDIM Inversion [197]	arXiv'24	LDM	Inversion	∼	–	Training-free
GeoNVS [28]	arXiv'24	VDM	Ptcloud	✓	–	Geo-grounded VDM
Proj. Repr. Cond. [56]	arXiv'24	LDM	3D proj.	✓	–	Feature warp
Novel View Extrap. [146]	arXiv'24	VDM	Cam traj.	∼	–	Large-angle extrap.
Reconstruction–diffusion hybrids						
GeNVS [23]	ICLR'23	LDM	NeRF feat	✓	NeRF	NeRF-cond diff.
ZeroNVS [144]	CVPR'24	LDM	Pose+int.	∼	–	In-the-wild
ReconFusion [183]	CVPR'24	LDM	pixelNeRF	✓	NeRF	Diff reg.
Cat3D [47]	NeurIPS'24	MV-Diff	Δpose	✓	3DGS	MV gen+recon
NVComposer [101]	arXiv'24	LDM	Feature	✓	–	Unposed multi-ref
Score distillation (text-to-3D)						
DreamFusion [13]	ICLR'23	LDM+NeRF	SDS	∼	NeRF	SDS
Magic3D [17]	CVPR'23	LDM+Mesh	SDS	∼	Mesh	C2F SDS
ProlificDreamer [175]	NeurIPS'23	LDM+NeRF	VSD	∼	NeRF	VSD
Fantasia3D [27]	ICCV'23	LDM+Mesh	SDS	∼	BRDF	Disentangled
GaussianDreamer [211]	CVPR'24	LDM+3DGS	SDS	∼	3DGS	SDS+3DGS
Taming SDS [170]	CVPR'24	LDM+NeRF	SDS	∼	NeRF	Mode collapse fix
Text-to-3D CSD [190]	ICLR'24	LDM+NeRF	CSD	∼	NeRF	Classifier SDS
Large-scale feed-forward reconstruction						
LRM [202]	ICLR'24	Transformer	Single	✓	Tripl.	Triplane LRM
TriplaneGaussian [222]	CVPR'24	Transformer	Single	✓	3DGS	Triplane+GS
GS-LRM [235]	ECCV'24	Transformer	Multi	✓	3DGS	Scalable LRM
Instant3D [96]	ICLR'24	LRM+MVDiff	Text	✓	3DGS	Text$\rightarrow$3D fast
CRM [176]	arXiv'24	Conv+LRM	Single	✓	Mesh	Conv reconstruct
High-res multi-view / production 3D						
ERA3D [98]	NeurIPS'24	LDM	Δpose	✓	–	Row-wise attn
MVDiffusion++ [141]	ECCV'24	LDM	Dense	✓	–	Dense HR MV
SplatFlow [52]	arXiv'24	RF-Diff	MV	✓	3DGS	Rectified flow
Hunyuan3D [189]	arXiv'24	LDM	Text/Img	✓	Mesh	Production 3D
Video diffusion 4-D generation						
SV3D [165]	ECCV'24	SVD	Orbit	✓	3DGS	3D from SVD
SV4D [194]	arXiv'24	SVD	Orbit+T	✓	4DGS	Dynamic 3D
CAT4D [182]	NeurIPS'24	VDM	View+T	✓	4D	4D from MV
4Diffusion [53]	arXiv'24	VDM	MV+T	✓	–	4D video diff

Table 5: World modeling and neural scene simulation methods.

Method	Year	3D Aware	Video	Action	Domain	Key Feature
GAIA-1 [61]	2023	∼	✓	✓	Driving	Seq. token WM
UniSim [223]	2023	✓	✓	✓	Driving	Sensor sim.
MARS [186]	2023	✓	✓	∼	Driving	Compositional
DriveDreamer [187]	2024	∼	✓	✓	Driving	Diffusion WM
OccWorld [166]	2024	✓	✓	✓	Driving	3D occupancy WM
WoVoGen [70]	2024	✓	✓	✓	Driving	World volume
Drive-WM [205]	2024	∼	✓	✓	Driving	MV forecasting
Genie [16]	2024	–	✓	✓	Games	Latent action
Genie 2 [123]	2024	✓	✓	✓	Open-world	Interactive 3D
WonderWorld [214]	2024	✓	–	–	Open-world	Instant 3DGS
ViewCrafter [215]	2024	✓	✓	–	Open-world	Cam-cond video
WorldDreamer [188]	2024	–	✓	∼	General	Masked token WM
AVID [116]	2024	–	✓	✓	General	VDM→WM
Cosmos [9]	2025	✓	✓	✓	Physical	WFM training
Street Gaussians [208]	2024	✓	✓	∼	Driving	3DGS driving
OmniRe [33]	2024	✓	✓	✓	Driving	All-in-one
DrivingGaussian [244]	2024	✓	✓	∼	Driving	Large-scale GS

Table 6: NVS vs. World Modeling capability boundary.

Capability	NVS (static)	World Model
Viewpoint-consistent rendering (D1)	✓	✓
Sparse/single-image input (D2)	∼	✓
Temporal dynamics (D3)	–	✓
Action/agent conditioning (D4)	–	✓
Physical plausibility (D5)	–	✓
Cross-scene generalisation	∼	✓
Causal scene understanding	–	✓

Table 7: NVS and world modeling datasets. **Type**: RT=real, Syn=synthetic. **Dyn.**: contains dynamic content. **Depth**: provides ground-truth or LiDAR depth. **Pose**: provides camera/LiDAR extrinsics. **Sem.**: provides semantic / instance labels. **Scale**: approximate number of scenes/objects/sequences. **Primary use**: main application in the NVS literature.

Dataset	Year	Type	Dyn.	Depth	Pose	Sem.	Scale	Primary Use
Object-centric								
ShapeNet [24]	2015	Syn	–	✓	✓	✓	51K obj.	Generative 3-D training
NeRF-Synthetic [11]	2020	Syn	–	✓	✓	–	8 obj.	Per-scene NeRF evaluation
Co3Dv2 [71]	2021	RT	∼	–	✓	✓	19K seq.	Category-level NVS
Objaverse [37]	2023	Syn	∼	✓	✓	∼	800K obj.	LRM / diffusion training
Objaverse-XL [38]	2024	Syn	∼	✓	✓	∼	10M+ obj.	Large-scale 3-D pretraining
MVImgNet [216]	2023	RT	–	–	✓	✓	238K seq.	Category generalizable NVS
Static indoor / outdoor scenes								
DTU [129]	2014	RT	–	✓	✓	–	124 scn.	MVS / NVS evaluation
LLFF [12]	2019	RT	–	–	✓	–	8 scn.	Forward-facing NeRF
Tanks&Temples [5]	2017	RT	–	✓	✓	–	14 scn.	Large-scale reconstruction
Mip-NeRF360 [73]	2022	RT	–	–	✓	–	9 scn.	Unbounded 360° NVS
ScanNet [3]	2017	RT	–	✓	✓	✓	1513 seq.	Indoor 3-D reconstruction
ScanNet++ [19]	2023	RT	–	✓	✓	✓	460 scn.	Generalizable NVS, hi-res
RealEstate10K [158]	2018	RT	–	–	✓	–	10K vid.	Generalizable video NVS
Dynamic scenes								
D-NeRF [6]	2021	Syn	✓	–	✓	–	6 obj.	Dynamic NeRF evaluation
HyperNeRF [79]	2021	RT	✓	–	✓	–	15 seq.	Topological deformation
CMU Panoptic [77]	2015	RT	✓	✓	✓	✓	65 seq.	Multi-view social activities
Neu3D [219]	2022	RT	✓	–	✓	–	6 seq.	Multi-view human video NeRF
iPhone Dataset [50]	2022	RT	✓	–	–	–	14 seq.	Casually-captured dynamic NVS
Autonomous driving & large-scale								
KITTI [4]	2013	RT	✓	✓	✓	✓	Urban	Early driving benchmark
Waymo Open [152]	2020	RT	✓	✓	✓	✓	1000 seg.	Driving sensor simulation
nuScenes [20]	2020	RT	✓	∼	✓	✓	1000 scn.	360° driving world model
KITTI-360 [203]	2022	RT	✓	✓	✓	✓	Urban	Urban scene NVS + panoptic
Argoverse 2 [179]	2023	RT	✓	✓	✓	✓	1000 seq.	High-res 360° driving
PandaSet [192]	2020	RT	✓	✓	✓	✓	103 scn.	LiDAR+camera NVS

Table 8: Quantitative NVS comparison on three standard benchmarks. NeRF-Synthetic: bounded synthetic objects (8 scenes). LLFF: real forward-facing scenes (8 scenes). Mip-NeRF360: real unbounded 360° scenes (9 scenes). Higher PSNR/SSIM and lower LPIPS are better. "–" indicates not reported on that benchmark. † denotes generalizable methods evaluated without per-scene fine-tuning.

Method	**Venue**	**NeRF-Synthetic**			**LLFF**			**Mip-NeRF360**		
		PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
Per-scene NeRF										
NeRF [11]	ECCV'20	31.01	0.947	0.081	26.50	0.811	0.250	–	–	–
Mip-NeRF [75]	ICCV'21	33.09	0.961	0.043	26.70	0.829	0.177	–	–	–
TensoRF [1]	ECCV'22	33.14	0.963	0.047	26.73	0.839	0.204	–	–	–
Instant-NGP [156]	TOG'22	33.18	0.963	0.037	25.59	0.817	0.224	–	–	–
Mip-NeRF360 [73]	CVPR'22	–	–	–	29.40	0.901	0.127	29.40	0.840	0.204
Zip-NeRF [74]	ICCV'23	–	–	–	30.38	0.921	0.100	30.38	0.866	0.179
Per-scene 3DGS										
3DGS [10]	TOG'23	33.32	0.963	0.044	27.21	0.815	0.183	27.21	0.815	0.183
Mip-Splatting [227]	CVPR'24	34.66	0.973	0.019	27.79	0.827	0.163	27.92	0.842	0.158
Scaffold-GS [113]	CVPR'24	33.56	0.966	0.033	27.63	0.827	0.176	27.71	0.831	0.171
2DGS [63]	SIGGRAPH'24	33.35	0.965	0.042	26.94	0.818	0.212	–	–	–
Generalizable (no per-scene training)†										
pixelNeRF [7]	CVPR'21	23.17	0.860	0.170	–	–	–	–	–	–
IBRNet [127]	CVPR'21	26.73	0.917	0.091	26.73	0.851	0.176	–	–	–
MVSNeRF [2]	ICCV'21	26.63	0.931	0.168	26.40	0.839	0.181	–	–	–
pixelSplat [35]	CVPR'24	–	–	–	26.31	0.815	0.200	–	–	–
MVSplat [32]	ECCV'24	–	–	–	26.40	0.823	0.188	–	–	–